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List of Abbreviations

2dfGRS	two degree field Galaxy Redshift Survey
2pcf	two-point correlation function
AGN	Active Galactic Nucleus
BOSS	Baryon Oscillation Spectroscopic Survey
CANDELS	Cosmic Assembly Near-infrared Deep Extragalactic Legacy Survey
CfA2	Center for Astrophysics 2
CMB	Cosmic Microwave Background
DEEP2	Deep Extragalactic Evolutionary Probe 2
EAGLE	Evolution and Assembly of GaLaxies and their Environments
GADGET	GAxaxies with Dark matter and Gas intEract
GAMA	Galaxy And Mass Assembly
GSMF	Galaxy Stellar Mass Function
HOD	Halo Occupation Distribution
ΛCDM	Lambda Cold Dark Matter
LSS	Large-Scale Structure
RSD	Redshift-Space Distortion
SDSS	Sloan Digital Sky Survey
sSFR	specific Star Formation Rate
SFR	Star Formation Rate
SFRD	Star Formation Rate Density
TNG	The Next Generation
zFOURGE	FourStar Galaxy Evolution Survey

Nomenclature

Symbol	Nomenclature	Value / Unit
c	speed of light	$2.99792458 \text{ m s}^{-1}$
h	dimensionless Hubble parameter	0.68
H_0	Hubble constant	$68 \text{ km s}^{-1} \text{ Mpc}^{-1}$
$L_{\odot}$	solar luminosity	$3.828 \times 10^{26} \text{ W}$ (or $\text{kg m}^2 \text{ s}^{-3}$)
$L_{\star}$	stellar luminosity	$\text{kg m}^2 \text{ s}^{-3}$
$M_{\odot}$	solar mass	$1.9884 \times 10^{30} \text{ kg}$
$M_{\star}$	stellar mass	kg
$\xi(r)$	(real-space) 2-point correlation function	—
r, r_p	distance between two galaxies	$h^{-1} \text{ Mpc}$
$w_p(r_p)$	projected 2-point correlation function	$h^{-1} \text{ Mpc}$
z	redshift	—

Abstract

Galaxy properties trace the local environment differently. Investigating this correlation provides valuable insights into the importance of considering environmental effects in galaxy formation and evolution models. Using the SIMBA cosmological hydrodynamical simulation and observational data, we measure and analyse galaxy clustering as a function of colour, specific star formation rate and age. The two-point correlation function of various galaxy samples is computed to analyse the clustering characteristics. The clustering is first measured in real space using the Landy-Szalay estimator. We also measure the projected correlation function $w_p(r_p)$ to compare with observations. To match BOSS (Baryon Oscillation Spectroscopic Survey) data, we create samples with a random selection based on probability distribution. A simple replication of the simulation box does not provide reliable results. Our results agree with the literature at small scales: redder, more quenched and older galaxies cluster more strongly. Those galaxies are more likely to be found in dense environments. We find that the simulation data alone predict the dependence on already mentioned properties using theoretical estimations in real space at $z = 0$. In redshift space at $z = 0.5$, our findings are consistent with previous studies, but only at small scales up to $r_p \sim 3 h^{-1}\text{Mpc}$, mainly because of the small size of the simulation box. At larger distances, the clustering of dark matter haloes significantly influences galaxy clustering. Due to poorly matched data, we do not obtain the clustering powers of the BOSS analysis. Nevertheless, our results still show the clustering dependence on the three properties. A larger simulation volume would be necessary for more accurate galaxy clustering measurements.

Key words: cosmology: large-scale structure — galaxies: evolution — galaxies: fundamental parameters — galaxies: haloes

Introduction

Galaxies are the building blocks of the Universe. They occur in various colours, sizes, and shapes, each property reflecting their physics of formation and evolution. Mappings from redshift surveys (e.g. SDSS, 2dfGRS) show the patterns of galaxies and matter on scales larger than individual galaxies (Peacock et al., 2002; Springel et al., 2006). Their spatial distribution forms groups or clusters. This distribution of visible matter in galaxies can be used to trace the underlying dark matter distribution. Modern cosmological models suggest that the visible baryonic matter is only a small component of matter in the Universe, the dominant element being dark matter.

The observed large-scale structure is thought to have emerged from the gravitational collapse of small-density perturbations from the early Universe. Small overdensities eventually grow and evolve into dark matter haloes through gravitational instability. Complex baryonic physics then happens to form stars and galaxies. Larger systems are, in turn, formed hierarchically. This is consistent with the structure formation scenario within the framework of the Λ CDM model, where low-mass galaxies formed first. High-mass galaxies only form later in the cosmic evolution (Shneider, 2015; Dodelson & Schmidt, 2021).

The dark matter halo properties correlate with both its environment and the properties of galaxies hosted by the halo. This prompts a dependence between galaxy properties and the environment (Wechsler & Tinker, 2018). Numerous observational studies also report that the large-scale distribution of galaxies is related to the properties of those galaxies. Brighter, massive, older, redder and quenched galaxies show a higher clustering strength than their fainter, less massive, younger, bluer and star-forming counterparts (e.g. Mostek et al., 2013; Skibba et al., 2014; Farrow et al., 2015; Durkalec et al., 2018; Zhai et al., 2023).

For the observational data to be effectively used but also to understand systematic effects, physical models of galaxy formation connecting galaxy properties to the evolving matter fields have to be set. Those models yield our current understanding theory of galaxy formation. For this purpose, various models have been adopted, namely semi-analytic or empirical models (halo occupation distribution,

sub-halo abundance matching) (e.g. Tinker et al., 2012; Campbell et al., 2018; Osato & Okumura, 2023). However, those approaches suffer some limitations as they are often based on dark matter-only simulations that solely focus on gravitational interaction. They do not predict a detailed clustering of baryonic matter, and some feedback effects are also left out, meaning that hydrodynamical processes are omitted. This is where hydrodynamical simulations play an important role (e.g. Vogelsberger et al., 2014; Schaye et al., 2015).

Though the small size of the simulation volumes was challenging at first, over recent years, promising results producing galaxy populations in reasonable agreement with observations have been made (for a review, see Crain & van de Voort, 2023). In particular, various hydrodynamical simulations of galaxy formations have been used to predict galaxy clustering: MassiveBlack-II¹ (Khandai et al., 2015), EAGLE² (Artale et al., 2017), IllustrisTNG³ (Springel et al., 2018), TNG100 and TNG300 (Einasto et al., 2023). Analysing the predicted properties, which can directly be obtained from the simulations, is traditionally known as part of exploring of such simulations. However, the limitations of the size of the simulation box tend to make clustering measurement less appealing. This makes the investigation of galaxy clustering based on their properties an interesting study to conduct using simulated galaxies.

This work aims to analyse the environmental dependence on galaxy properties using a representative mock Universe. The clustering is measured for galaxy samples selected by three properties: the colour, the specific star formation rate (sSRF) and the age.

This thesis is organized as follows. The first chapter introduces the basic concepts we need to understand to conduct this study. The second chapter describes the data and methods. The third chapter presents and discusses the results.

¹ <http://mbii.phys.cmu.edu>

² <https://eagle.strw.leidenuniv.nl/>

³ <https://www.tng-project.org/>

Chapter 1

Overview

1.1 Parameters characterizing a galaxy

The galaxy population is very diverse. To characterize a galaxy, a great number of parameters are necessary given that galaxies are a diverse class of objects. Several parameters are correlated with each other. Galaxy formation and evolution is a complex subject to study mainly because time and space scales are tremendously larger. Nevertheless, comparing galaxy properties at different epochs can give insights into their evolution. Important information can also be obtained regarding their underlying physical process, which can lead to a model of their formation. This section aims to introduce some observational aspects commonly applied to study galaxies: morphology, luminosity, colour, stellar mass, star formation rate, environment and nuclear activity.

1.1.1 Morphological classification

Searching for patterns and features in observations, which is a purely empirical method, allows for a classification. In this context, Edwin Hubble, in 1936, classified galaxies according to their appearance in optical wavelengths (Hubble, 1936). This classification is based on morphology and informs about how stars, gas and dust are distributed. The four main types are: elliptical, lenticular, spiral, and irregular, each type has its own specific physical properties as shown in Figure 1.1.

Ellipticals (E) are usually smooth and round without any clearly defined structure. Their smooth appearance suggests their stars have had time to achieve a well-mixed equilibrium state. This corresponds to an old metal-rich stellar population reflecting an early intense process of star formation. They contain, in most cases, a lower mass fraction of gas and dust. Ellipticals are subdivided according to their eccentricity and denoted En where $E0$ looks circular whereas $E6$ looks highly eccentric.

Spiral galaxies (S) consist of a central bulge that looks elliptical and a disk with a spiral structure

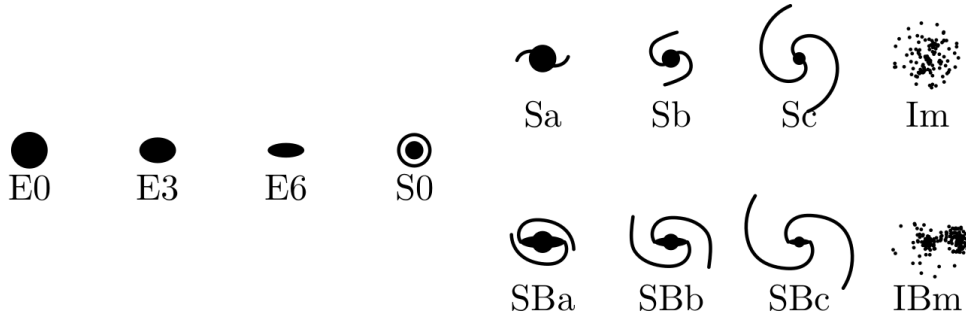


Figure 1.1: A schematic representation of the “tuning fork” Hubble sequence showing the morphological classification of galaxies according to Abraham (1998).

extended over a large distance. The spiral arms are a striking characteristic of spirals containing a high density of gas (cool gas HI, molecular hydrogen HII) and dust, indicative of active star formation. They are divided into normal (*S*) and barred spirals (*SB*). The bar is present in the central part for about more than half of disk galaxies. They are sorted into subclasses from *a* to *d* according to three criteria: the light fraction in the central bulge, the tightness of the spiral arms and the level to which the arms are resolved into stars, HII regions and ordered dust lanes. They are denoted *Sa*, *SBa*, *Sb*, *SBb*...

Lenticular galaxies (*S0*) are an intermediate class between ellipticals and spirals. They contain a central elliptical bulge but with a region that appears like a disk lacking spiral arms. As for irregular galaxies (*Irr*), they do not display any regular defined structure.

The Hubble sequence classifies galaxies according to their mix of the disk and ellipsoidal components, ranging them from early-type (pure ellipsoid) to late-type (pure disk). This is only a limited classification based on observed morphology. Indeed, the problems with this classification include the projection effects, the wavelength used for the observation, or even the fact that some galaxies do not fit into any of the classes. However, the sequence holds an important aspect, which is that many intrinsic properties like colour or star formation rate, along with some mechanisms of formation, change systematically along the sequence (Mo et al., 2010; Jain, 2015; Shneider, 2015).

1.1.2 Galaxy properties

Luminosity and colour: The luminosity of a galaxy is one of its most basic properties. It is proportional to the number of stars in the galaxy. It can change over time following the formation and evolution of the stars. Galaxies occur in a broad range of luminosity, with the brightest reaching $\sim 10^{12}L_{\odot}$, where $L_{\odot}$ is the luminosity of the Sun. The faintest galaxy is, however, not well-defined as it changes regularly according to new discoveries.

The colour is defined as the luminosity ratio measured through two different photometric

passbands. A red galaxy has a relatively higher luminosity in the redder passband than in the bluer passband. Early-type galaxies generally show a redder colour than the late-type ones. The colour of a galaxy is also related to the age and metallicity of its stellar population. Redder galaxies are generally older and/or more metal-rich (Mo et al., 2010; Ferreras, 2019).

Stellar mass: Another basic galaxy property is the stellar mass which is also related to the total number of stars. It is measured in units of solar mass. For early-type galaxies, the stellar mass varies over a wide range of values ($10^7 M_\odot \lesssim M_\odot \lesssim 10^{14} M_\odot$), whereas late-type galaxies cover a smaller range in mass limited to $10^9 M_\odot \lesssim M_\odot \lesssim 10^{12} M_\odot$ (Shneider, 2015).

A fundamental cosmological observable in the study of statistical properties of galaxies is the galaxy stellar mass function (GSMF), defined as the number density of galaxies in bins of stellar mass at a given redshift z . The shape and the evolution of the GSMF inform about the growth of the baryonic content of the Universe. They also can be used to infer the star formation activity and the presence of cold molecular gas across cosmic time (Weaver et al., 2023).

Star formation rate (SFR): It is defined as the mass of stars that form per unit time in a galaxy, typically measured in units of $M_\odot \text{ yr}^{-1}$. When the SFR is measured per unit comoving volume, it is called star formation rate density (SFRD) and expressed in units of $M_\odot \text{ yr}^{-1} \text{ Mpc}^{-3}$. The specific SFR (sSFR) is defined as the SFR divided by the stellar mass M_* . Based on observations, it is generally accepted that the overall SFR of a galaxy is determined by its ability to form dense molecular clouds that also contain large amounts of dust. There is a strong correlation between the restframe colour and the SFR because the latter reflects the star formation history of the stellar population within a galaxy (Shneider, 2015; Ferreras, 2019).

1.1.3 Environment

Environment refers to the large-scale region where galaxies are located and can be described by the density averaged over volumes that include many galaxies. This gives a connection between the local formation process in galaxies to the large-scale structure (LSS) of the Universe. Observations on large scales indicate that the spatial distribution of galaxies shows a variety of structures. Galaxies are located in different environments, some are found in high-density clusters containing thousands of galaxies. Some others are located in groups containing ten to hundreds of galaxies, while others are distributed in low-density filaments or sheets. Some regions are even called voids and contain no bright galaxies. All of these are seen in Figure 1.2. This location in the LSS affects galaxy formation

and evolution as there are important environment-related processes (mergers, ram pressure stripping or harassment) (Mo et al., 2010; Shneider, 2015; Ferreras, 2019).

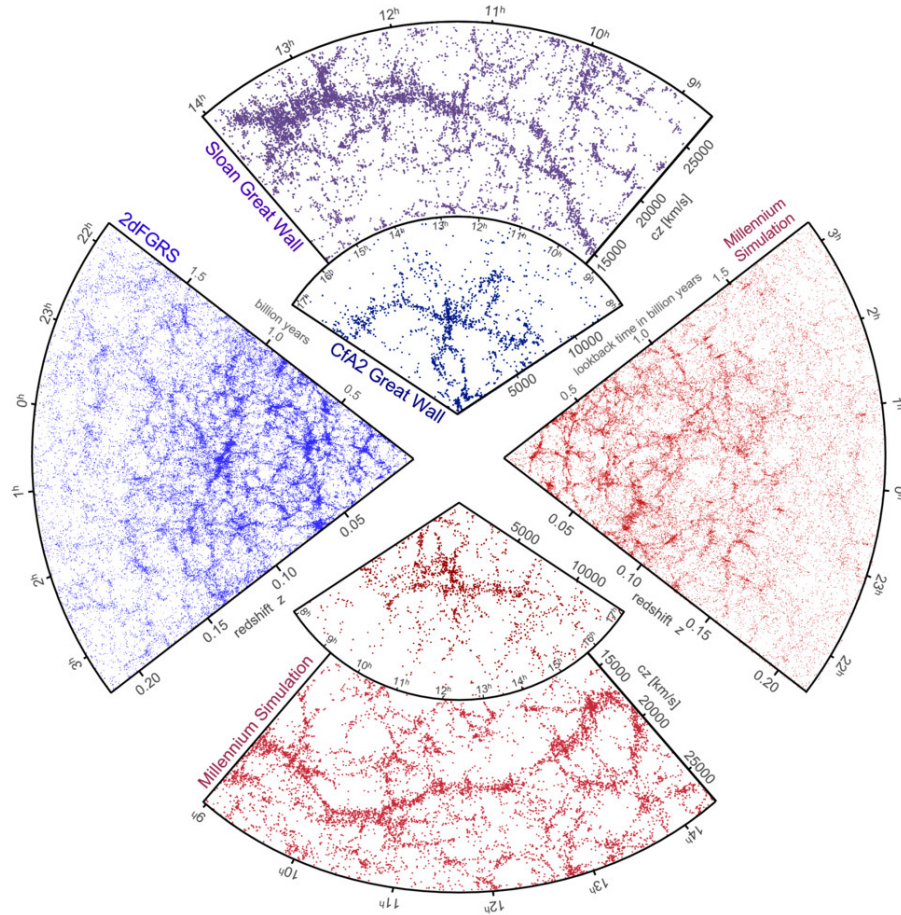


Figure 1.2: The galaxy distribution showing the “cosmic web” in various spectroscopic redshift surveys and in mock catalogues constructed from cosmological simulations. The small slice at the top shows the CfA2 “Great Wall”, with the Coma cluster at the centre. Drawn to the same scale is a small section of the SDSS, in which the “Sloan Great Wall” is visible. The wedge on the left shows one-half of the 2dFGRS. At the bottom and on the right, mock galaxy surveys constructed using semi-analytic techniques to simulate the formation and evolution of galaxies within evolving dark matter distribution of the Millennium simulation are shown, selected with matching survey geometries and magnitude limits (Springel et al., 2006).

Early studies in the 1930s first showed that the morphological mix of galaxies depends on the environment. High-density environments host a larger fraction of early-type galaxies. Over time, recent studies backed up this feature and added more details into the environmental dependence on galaxy properties. A denser environment not only hosts more early-type galaxies but also those that are, on average, more massive, redder, more concentrated, less gas-rich and with a lower SFR. However, it is challenging to interpret and draw conclusions in terms of galaxy formation and evolution due to the fact that various galaxy properties are strongly correlated, even if the environment is fixed.

1.1.4 Nuclear activity

Most galaxies host a supermassive black hole at their centre. However, only a small fraction of those galaxies are called active. Gas flowing into the black hole forms an accretion disk where the matter gets heated. The accretion emits large amounts of radiation in a broad range of wavelengths and produces an active galactic nucleus (AGN) outshining the host galaxy.

AGN activity affects the flow and the heating of the gas that fuels star formation in galaxies. In addition, nuclear activity has a feedback mechanism that contributes to the quenching of star formation in massive galaxies. Another aspect of the impact of the black hole on the host galaxy is the strong correlation between galaxy properties and the mass of the central black hole. Therefore, the presence of the black hole is strongly related to galaxy evolution (Mo et al., 2010; Ferreras, 2019; Fontanot et al., 2020).

1.2 Large-Scale Structure

On sufficiently large scales, the Universe is considered to be spatially homogeneous and isotropic. This hypothesis is known as the cosmological principle and is one of the bases of modern cosmology. However, on small scales, less than ~ 100 Mpc, the Universe is clearly inhomogeneous, as galaxy distribution from redshift surveys has shown (Figure 1.2). The latter supplies the most direct measurements of the galaxy density field, from which it is possible to obtain their positions and redshifts that can be explored in terms of three-dimensional statistics. The statistical properties of this density field characterize the distribution of galaxies in space. This provides insights into the clustering of galaxies, which is an important study for understanding galaxy formation since it may provide clues about how galaxies influence each other as they form and evolve (Mo et al., 2010; Shneider, 2015; Sureshkumar, 2022).

1.2.1 Correlation function

As already mentioned, the way that galaxies are structured together indicates that their distribution is not random. Instead, they tend to be grouped together. In other words, the probability of finding a galaxy at a position $\mathbf{x}$ depends on whether there is another galaxy located at a position $\mathbf{y}$ in its vicinity. Thus, it is more likely to find a galaxy close to another one than at an arbitrary position.

Let us consider the distribution of galaxies to have an average number density $\bar{n}$ with two points in space $\mathbf{x}$ and $\mathbf{y}$ and two volume elements dV around them. The probability of finding a galaxy in the volume element dV around $\mathbf{x}$ (or $\mathbf{y}$) is given by Equation (1.1).

$$P_1 = \bar{n} dV \quad (1.1)$$

Because of the existing correlation in the distribution of galaxies, the joint probability of finding a galaxy in dV element at position $\mathbf{x}$ and finding a galaxy in dV element at position $\mathbf{y}$ is given by Equation (1.2).

$$P_2 = (\bar{n} dV)^2 [1 + \xi(r)] \quad (1.2)$$

$\xi(r)$ is defined as the galaxy two-point correlation function (2pcf), where r is the distance between $\mathbf{x}$ and $\mathbf{y}$ (Peebles, 1980). A random galaxy distribution would give $\xi(r) = 0$ since this means Equation (1.2) would be $P_2 = P_1^2$, meaning that there is no correlation. Hence, 2pcf defines the excess probability of finding a galaxy at a distance r from another galaxy, relative to a random distribution. In a measurement of 2pcf, any deviation of $\xi(r)$ from zero indicates spatial clustering.

Another statistical tool that can be used to analyse the galaxy density field is the power spectrum. Related by a Fourier transform, the power spectrum and the correlation function contain the same information. Both provide useful information about cosmic matter distribution based on galaxy clustering. The choice of the method depends on convenience. But we will see later in this chapter that when it comes to galaxy properties, the correlation function is what is traditionally used.

1.2.2 Redshift-space distortion

In 1928, Hubble discovered that the radial velocity r of galaxies, which is measured with Doppler shift of spectral lines, and the distance D of galaxies are linearly connected by the Hubble law defined by Equation (1.3):

$$r = H_0 D \quad (1.3)$$

where the constant of proportionality H_0 is called the Hubble constant. Galaxies seem to be moving away from us with a velocity that is linearly increasing with their distance reflecting the expansion of the Universe known as the Hubble expansion. However, cosmic sources do not exclusively follow the Hubble expansion; deviations are caused by local gravitational fields, which are the results of local density fluctuations. The result is an acceleration that affects the matter and creates peculiar velocities. Most galaxies are not stationary; they have non-negligible peculiar velocities.

Because of these peculiar motions, the observed redshift of a galaxy is the superposition of the cosmic expansion velocity (cosmological redshift) and its peculiar velocity v along the line of sight; the other two spatial coordinates (angular coordinates) are not impacted by the peculiar velocity.

Consequently, a distortion of the galaxy positions is seen in wedge diagrams, indicating a shift in the radial direction with respect to their true positions.

In fact, it is challenging to obtain accurate distances due to difficulties in observational studies. Instead, the redshift of galaxies is quite easy to get. Using redshift as a distance, the inferred distance s to a galaxy is related to its redshift z by $s = cz$, where c is the speed of light. Therefore, the observed three-dimensional position of a galaxy is defined by the angular coordinates and the redshift distance.

$$s = r + v \quad \implies \quad cz = H_0 D + v \quad (1.4)$$

The distortions caused by the peculiar velocities contain additional information about the matter density field. This is because of the existing correlation between the mass distribution and the galaxy positions. Therefore, there are modifications of the galaxy statistics which are known as redshift-space distortions (RSDs) (Mo et al., 2010; Shneider, 2015; Dodelson & Schmidt, 2021).

1.2.3 Dark matter haloes

In the current theory of the formation and evolution of cosmic structure, $\sim 84\%$ of the mass in the Universe is made of dark matter (Planck Collaboration et al., 2020). Modern cosmological models establish the evidence for its existence through a wide range of observations on all scales (galaxies, galaxy clusters, LSS, CMB) (Freese, 2009). In this context, dark matter haloes are the basic unit into which matter collapsed. They can be described as gravitationally bound and virialized structures formed in the evolution of the cosmic density field.

The formation of the haloes began during cosmic inflation, where density perturbations were generated in the first fraction of a second. Then, gravitational instability grew these perturbations over time. They had grown linearly until reaching a critical density, after which they decoupled from the expansion of the Universe and collapse to form virialized dark matter haloes. These haloes continued to grow in mass and size, either by accreting material from their neighbourhood or by merging. At first, gas and dark matter were well mixed but as the Universe evolved, gas began to dissipate and fell into the dark matter haloes. For large enough haloes, gas was able to cool and form stars and galaxies hierarchically (Mo et al., 2010; Shneider, 2015; Wechsler & Tinker, 2018).

Since galaxies reside in dark matter haloes, there is a tight correlation between halo properties and galaxy properties. This physical and statistical connection is often referred to as the galaxy-halo connection (Wechsler & Tinker, 2018). For instance, if we generally assume that the basic properties of the haloes are known for a given cosmological model, the galaxy-halo connection can be described by an empirical model known as halo occupation distribution (HOD). It specifies the probability

distribution for the number of galaxies meeting some criteria (a stellar mass threshold for instance) in a halo, which is generally determined by the halo mass (e.g. Berlind & Weinberg, 2002).

1.3 Environmental dependence on galaxy properties

The measurement of galaxy clustering and its correlation with galaxy properties have been studied in observations (e.g. Zehavi et al., 2011; Mostek et al., 2013; Skibba et al., 2014; Farrow et al., 2015; Durkalec et al., 2018; Zhai et al., 2023). These investigations hold important information in terms of galaxy formation and evolution. The main conclusion from observations is that brighter, massive, redder, older galaxies are more strongly clustered than their fainter, less massive, bluer, younger counterparts. This means that galaxies showing a stronger clustering are more likely to be found in high-density environments in the large-scale structure of the Universe. Therefore, various galaxy properties influence galaxy clustering on small scales differently. That is the main objective of investigating the environmental dependence on galaxy properties.

The galaxy 2pcf is traditionally used for this kind of study. But other tools have been used before, including luminosity functions or galaxy local density (e.g. Zucca et al., 2009; Cucciati et al., 2006, 2010). The conclusions from those studies confirm that galaxy properties trace the local environment differently.

1.4 Galaxy formation and evolution in numerical simulations

It is widely acknowledged that the collapse of gas leads to the formation of disk galaxies, while ellipticals are mainly formed by interactions and mergers. Nevertheless, the formation of galaxies is strongly governed by complex baryonic processes. The interconnected processes include gas accretion, star formation, chemical enrichment and large-scale outflows (Wright et al., 2024). Additionally, the dynamic range in length scales involved in galaxy evolution within a cosmological context is very large. The fact that the evolution of galaxies is related to the local environment makes it important to consider a sufficiently large volume of the Universe. But star formation in galaxies occurs in much smaller sizes. The dynamic range would then be at least 10^8 in length (Shneider, 2015).

Modern cosmological hydrodynamical simulations (Vogelsberger et al. 2014; Schaye et al. 2015; Springel et al. 2018; Davé et al. 2019) incorporate those baryonic physics, making such simulations powerful tools to address numerous cosmological questions, including galaxy formation and evolution. To get accurate results, the simulations must satisfy the following criteria (Khandai et al., 2015):

- a large enough simulation volume that is a representative region of the Universe, because otherwise clustering and mass function inferences would not be correct,
- high enough mass and spatial resolution such that the properties of the relevant objects have converged, meaning that the number of particles included in the simulations must be high enough, depending on the case.

The outputs of numerical simulations should be robust to changes in various aspects, like the resolution or the simulation volume. This characteristic is known as convergence. It is a way to ensure that the output data are valid. Galaxy populations from different simulation volumes and resolutions can be compared to analyse the convergence (Crain & van de Voort, 2023).

Chapter 2

Methodology

2.1 SIMBA simulation and data

Data from the SIMBA¹ cosmological hydrodynamical simulation are used to conduct our study. Introduced in Davé et al. (2019), SIMBA is a state-of-the-art suite of galaxy formation simulations for exploring the co-evolution of galaxies, black holes and intergalactic gas within a cosmological context. SIMBA is the descendant of the MUFASA simulation (Davé et al., 2016). Both use a modified version of the GIZMO² cosmological gravity plus hydrodynamics solver (Hopkins, 2015). This code is built upon GADGET-3³ galaxy formation code (Springel, 2005) and evolves dark matter and gas elements together with gravity and pressure forces.

To quench galaxies, MUFASA’s prescription is the implementation of a heuristic quenching feedback in which diffuse gas within massive haloes was prevented from cooling. However, this energy is considered as provided by Active Galactic Nuclei (AGN) accretion and the interaction of the black hole energy release with surrounding gas. This lack of black hole growth and AGN feedback depiction is where MUFASA was physically less well-motivated. Conversely, SIMBA employs multi-scale prescriptions for key physical processes governing galaxy evolution, including modelling black hole growth. It uses much of the framework of MUFASA with various updates following recent theoretical and observational results. The most significant change is that black holes are seeded and grown during the simulation with the accretion energy as a feedback driver for quenching galaxies. Those features are crucial since, as already mentioned in Chapter 1, they are tightly connected to galaxy evolution.

¹ <http://simba.roe.ac.uk/>

² <http://www.tapir.caltech.edu/~phopkins/Site/GIZMO.html>

³ <https://wwwmpa.mpa-garching.mpg.de/gadget/>

Performance of the model

To ascertain SIMBA is well-suited for this work, we assess the performance of the model by ensuring that SIMBA reproduces observed stellar and gas properties of the galaxies. To that end, SIMBA's predictions are compared to results from observational data and other simulations. If it fits, the model produces a good simulation and is a robust platform to study galaxy evolution in detail.

The galaxy stellar mass function (GSMF) (Figure 2.1) is reasonably well measured over much of cosmic time. This is among the reasons why it represents a tight test for fundamental feedback modules of a galaxy formation model.

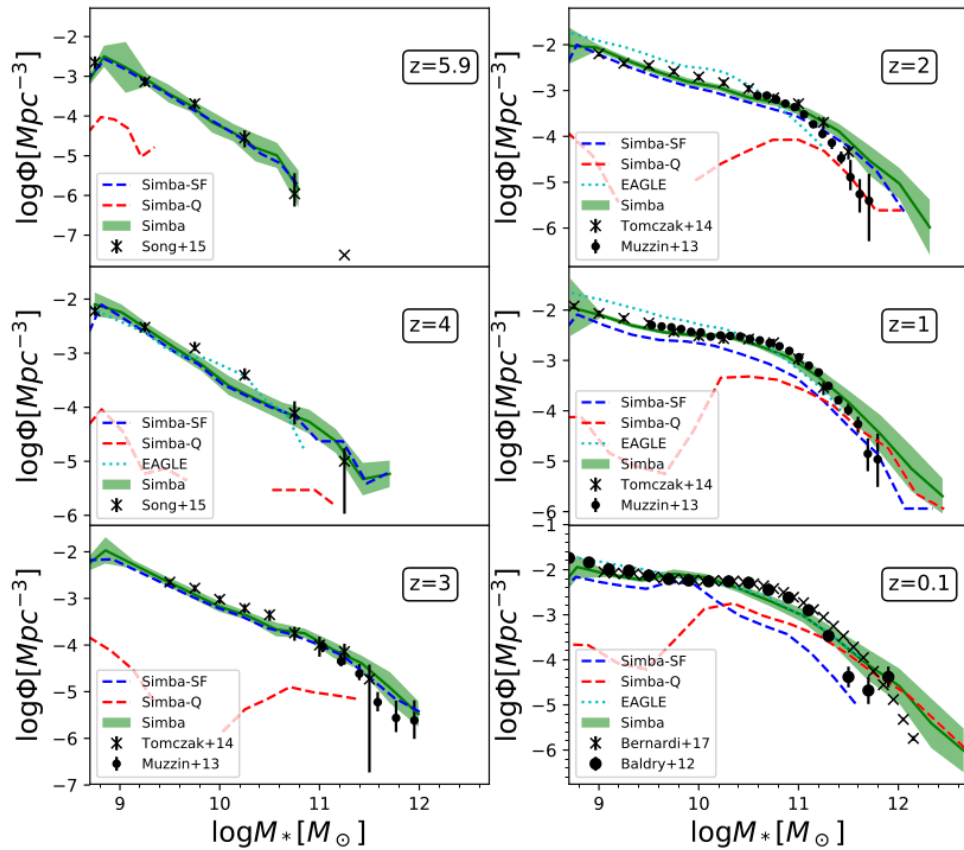


Figure 2.1: Stellar mass function evolution from $z = 6 \rightarrow 0$, compared to observations (at $z = 0$ from Bernardi et al. (2017); at $z = 1, 2, 3$, from Tomczak et al. (2014) combining CANDELS and zFOURGE data; and at $z = 4, 6$, from Song et al. (2016) based on CANDELS) as indicated in the legends. The green bands show the results from all SIMBA galaxies, with the spread computed from jackknife resampling the eight simulations sub-octants. The red and blue dashed lines show the mass functions of central galaxies subdivided into quenched (Q) and star-forming (SF) samples respectively at $s\text{SFR} = 10^{-1.8+0.3z} \text{ Gyr}^{-1}$. The cyan-dotted lines show the results from EAGLE for comparison (Davé et al., 2019).

The GSMF characterizes how efficient are haloes at converting their baryons into stars. Figure 2.1 shows the GSMF at $z = 0.1, 1, 2, 3, 4, 6$ from SIMBA (green lines). In general, within systematic

uncertainties, SIMBA produces good agreement with the observed GSMFs at all redshifts. And compared to the EAGLE simulation, the results are overall in quite great agreement as well (Davé et al., 2019).

Additionally, another key test for cosmological galaxy formation models that has long been used is the evolution of the cosmic SFR density (SFRD). Figure 2.2 shows the cosmic SFRD across cosmic time in SIMBA compared to Madau & Dickinson (2014) compilation which is a homogenized database for SFRD. SIMBA’s peak is slightly shifted towards higher redshift and also slightly lower than observed, meaning that stars are formed earlier in SIMBA. Nevertheless, the overall shape and amplitude are in very good agreement with observations and comparable to another simulation, IllustrisTNG (Pillepich et al., 2018).

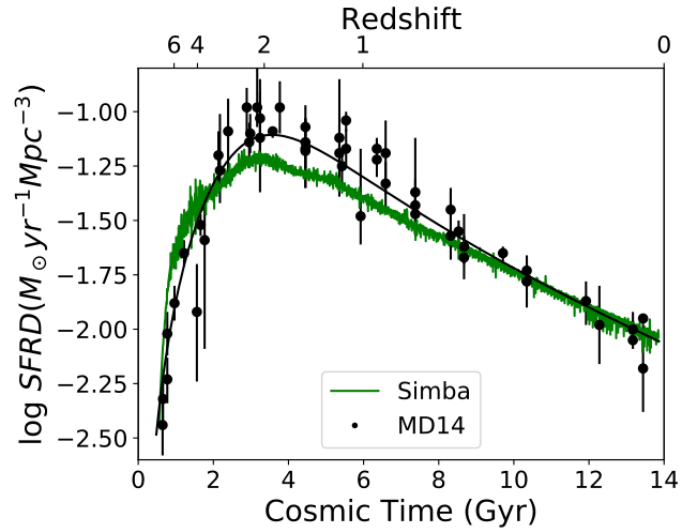


Figure 2.2: Star formation rate density evolution versus age of the Universe in SIMBA (curve), compared to the observational compilation of Madau & Dickinson (2014) (black points and best-fitting line) (Davé et al., 2019).

Apart from this, other predictions of the model are given below.

The star-forming main sequence of galaxies in SIMBA shows a good agreement with observations at $z = 0$. It is also similar to the trend seen in EAGLE indicating a tight connection between quenching and black holes. Moreover, SIMBA gas fractions, neutral and molecular, show a dropping trend with M_* which is consistent with observations. Also, more gas-rich galaxies at a given M_* seem to have a higher SFR. This correlation is fairly strong in MUFASA, a trend that is also seen in EAGLE and in observations.

These are only some examples of the performance to validate the model. But as detailed in Davé et al. (2019), the results demonstrate that SIMBA broadly reproduces a wide range of observed galaxy

properties (such as M_* , SFR or gas content). Though some aspects are in conflict with observations, SIMBA provides a state-of-the-art platform for studying galaxy evolution across a range of masses, environments and cosmic epochs.

Simulated data

The primary SIMBA runs contain 1024^3 dark matter particles and 1024^3 gas elements in a $100 h^{-1}\text{Mpc}$ volume ($h = H_0/100$). It assumes a cosmology described in Planck Collaboration et al. (2016). Here, we mainly use a smaller box of $50 h^{-1}\text{Mpc}$ with 512^3 dark matter particles and 512^3 gas elements. We also consider a $25 h^{-1}\text{Mpc}$ volume with the same number of particles, hence a higher resolution.

Galaxies and haloes are cross-matched in post-processing using CAESAR⁴ python package which is an YT⁵ extension; the latter being another python package for analysing and visualizing volumetric data. CAESAR was developed to specifically analyse outputs from cosmological simulations. Taking a SIMBA snapshot as input, CAESAR outputs a compact and portable hdf5 catalogue that contains all galaxy and halo information with various pre-computed key properties. Particle lists of individual galaxies and haloes are also available in a way that other properties can be obtained with python scripts (Davé et al., 2019). The data can then be explored without the original simulation binary.

SIMBA snapshots and galaxy catalogues are available for each of the 151 output redshifts from $z = 20$ to $z = 0$. The CAESAR galaxy catalogues that align with our work are detailed in Table 2.1.

Table 2.1: SIMBA data (CAESAR catalogue) that are used.

Name	Snapshot number	Box size [$h^{-1}\text{Mpc}$]	Redshift z
m25n512	125	16.77362	0.49043
m50n512	151	49.99999	0
m50n512	125	33.54724	0.49043
m50n512	122	32.11634	0.55683
m100n1024	125	67.09449	0.49043

2.2 Sample selection

To assess the clustering dependence on galaxy properties, we create various galaxy samples. The properties that we choose in this work are: r–i colour, specific star formation rate (sSFR) and age.

⁴ <https://caesar.readthedocs.io/en/latest/>

⁵ <https://yt-project.org/>

But the study can be extended to other properties as well.

We divide SIMBA galaxies in four bins of colour, sSFR and age, creating red and blue, quenched and star-forming, old and young subsamples along with the intermediate bins if necessary. Each one contains 25% of the galaxies from the original full sample.

2.2.1 Interpolation

For our results to be comparable to Zhai et al. (2023), we match our SIMBA galaxies to BOSS galaxies by referring to the stellar mass function in Appendix E of Maraston et al. (2013). There are three redshift bins in the mass function, we choose the closest to the analysis in Zhai et al. (2023). The resolution needs to be taken into account, in addition to the redshift. The bin with the best resolution is the first one, $z = 0.45 - 0.55$, which falls within the BOSS data we are referring to ($0.4 < z < 0.62$). The second bin, $z = 0.5 - 0.6$, is also good in terms of redshift range, but it has a lower resolution. Hence, we take the first redshift bin. However, we still tested with the second bin in Appendix A and found that overall there is no significant difference (Figure A.1).

The method that we adopt is as follows. A sample is randomly selected based on probability distribution. We take the redshift bin $z = 0.45 - 0.55$ (Appendix E in Maraston et al. 2013) as a reference. The corresponding stellar mass is discontinuous, so to map the SIMBA galaxy stellar mass to the probabilities given by the BOSS stellar masses, we need to interpolate.

The steps are given below:

- get the original distribution of stellar mass from BOSS galaxies,
- compute the normalised cumulative sum, which will serve as the cumulative distribution function (CDF),
- create an interpolation function for the CDF using `interp1d` from `scipy.interpolate`,
- get the probability density function (PDF) by differentiating consecutive values of the CDF,
- create an interpolation function for the PDF,
- get a new sample with SIMBA galaxy stellar masses, where the probability is 0 for anything outside the BOSS range. We choose SIMBA galaxy catalogue corresponding to $z = 0.5$,
- get the probabilities (PDF) for each point in that new sample,
- randomly select galaxies based on the defined probabilities to create a new galaxy sample.

Figure 2.3 shows the stellar mass distributions before and after the interpolation (with different sample sizes). We can see that SIMBA’s distribution is approaching BOSS’s in Maraston et al. (2013) after the interpolation which is expected.

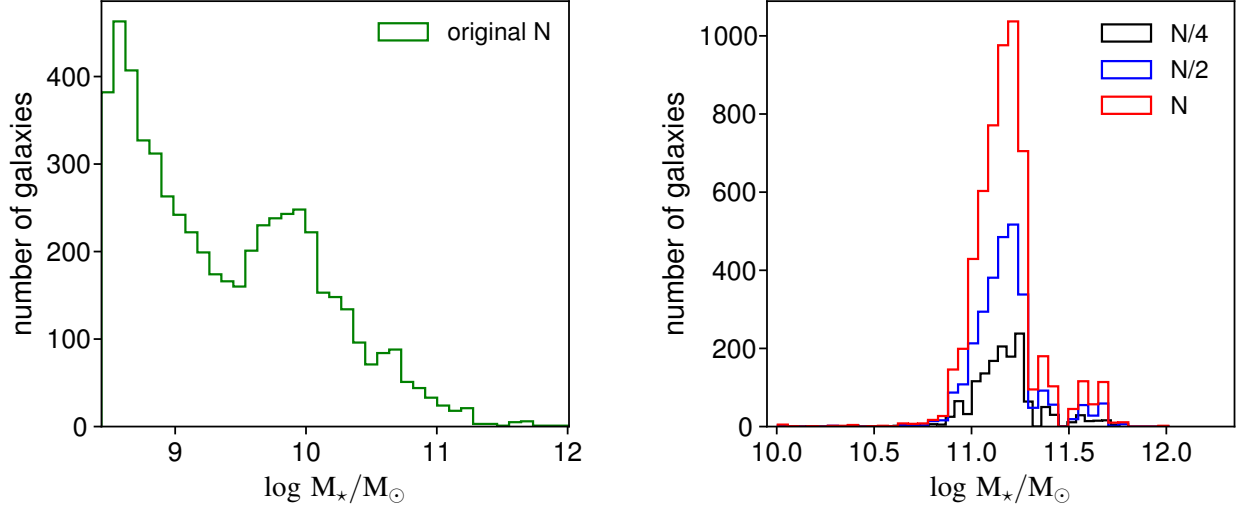


Figure 2.3: Stellar mass distributions of SIMBA galaxies at $z = 0.5$ before (*left*) and after (*right*) the interpolation. The original distribution is displayed in green. The new distributions show three different samples: full sample (red), full sample divided by 2 (blue) and full sample divided by 4 (black).

Effect of the interpolation

To assess how the new samples could affect our results, we analyse the resolution. The first thing to do is to evaluate the change of clustering by gradually increasing the resolution. In that case, four samples determined by the lowest stellar mass ($\log M_*/M_\odot > 9 - 9.5 - 10 - 10.5$) are used and we measure the clustering on each sample.

In addition, the clustering change can be assessed by using the BOSS probability distribution applied to SIMBA with different sample sizes. To do this, the data require an oversampling because when applying the probability distribution, we get duplicate elements in the samples. Therefore, we add random values to the positions before the measurement. Those random values are drawn from a Gaussian distribution centred at 0 with a standard deviation of δ , where δ is the average distance between the galaxies in the box and should be in the range of $[\text{boxsize}/N_{\text{galaxies}}, \text{boxsize}/2]$. The aim is to smooth the distribution. Hence, the overall distribution (of the positions) should not change.

To find the adequate value, we vary δ and see how this affects both the distribution of the positions and the clustering. Statistical tests are performed using `ttest_ind` and `ks_2samp` from `scipy.stats` to find p-values so that we can compare the distributions to the original ones. The objective is to find the δ such that the distributions of the positions are still drawn from the same population as the original distributions.

2.2.2 Duplication of the simulation box

Another approach to match SIMBA data with BOSS observations is to duplicate the original simulation box. Instead of an interpolation based on probability distribution of stellar masses from observations (section 2.2.1), we simply replicate the $50 h^{-1}\text{Mpc}$ simulation box to reach a reasonable number of galaxies. Referring to the mass distribution of BOSS galaxies in Maraston et al. (2013), we retain the galaxies within the range where the peak of the mass distribution is located. Beyond that range, galaxies are randomly dropped. The data are summarised in Table 2.2.

Table 2.2: Number of galaxies in BOSS and SIMBA original box at selected redshifts.

Data	N_{gal}	Redshift
BOSS (Zhai et al., 2023)	209 697	$0.48 < z < 0.62$
BOSS (Maraston et al., 2013)	132 693	$0.45 < z < 0.55$
SIMBA	5 685	$z = 0.5$

The peak of masses of observed galaxies is $\sim 10^{11.15}M_{\odot}$ with 26446 galaxies. But the corresponding number of simulated galaxies is only 23 according to the mass distributions detailed in Table C.1 of Appendix C. Therefore, the original simulation box should be replicated about 9 or 10 times on each side of the box.

It is not necessary to duplicate the galaxies in the box, we only need their position. To obtain the replicated positions in the larger box, we proceed as follows:

- define the number of replications along each dimension,
- create the offsets for each dimension,
- generate the offsets in three dimensions,
- replicate the positions of the galaxies in the original smaller box for each offset and add the offsets to create the larger box.

2.3 Galaxy clustering measurement

2.3.1 Two-point correlation function

In this section, we describe the methods we employ to measure and analyse the clustering throughout this study, which are based on the methodology in Zhai et al. (2023) that aligns with our work.

As mentioned in section 1.2.1, the two-point correlation function (2pcf) $\xi(r)$ statistically describes the spatial distribution of a given galaxy sample. It can be used to measure the clustering of galaxies (Peebles, 1980). In practice, 2pcf is computed using estimators.

2pcf estimators

The 2pcf measurement is affected by the limited survey areas or simulation volumes in practice. As a result, several estimators, all based on pair counting, have been used in the literature to measure $\xi(r)$ to minimise those effects. Three of the most well-known of them are defined below.

$$\xi_{DP}(r) = \frac{DD}{RR} - 1 \quad (2.1)$$

$$\xi_H(r) = \frac{DD \cdot RR}{DR^2} - 1 \quad (2.2)$$

$$\xi_{LS}(r) = \frac{DD - 2DR + RR}{RR} \quad (2.3)$$

where DD is the normalised observed number of galaxy-galaxy pairs separated by distance r in the real galaxy sample. RR is similar to DD but from the random sample. And DR is the normalised number of cross pairs of galaxies between the real and random samples. Equations (2.1), (2.2) and (2.3) were respectively introduced by Davis & Peebles (1983), Landy & Szalay (1993) and Hamilton (1993).

To know which of these estimators to use, they have been compared. The comparison criterion considers bias and variance and is independent of the possibly non-Gaussian nature of the error statistics (Kerscher et al., 2000). The estimator that seems to work better than others is the Landy-Szalay estimator, as it particularly reduces the cosmic bias caused by edge effects at large distances. But the performance is almost the same as with the Hamilton estimator. Apart from this, Sureshkumar (2022) measured the clustering of a sample using the three estimators and found that $\xi_{LS}(r)$ is less noisy than $\xi_{DP}(r)$ and $\xi_H(r)$.

In our case, we use the Landy-Szalay estimator (Landy & Szalay, 1993) defined by Equation (2.3), which was also used in Zhai et al. (2023).

DD , DR and RR are calculated as follows.

- Extract the relevant objects, galaxies in this case, from the CAESAR file,
- Pair the galaxies and knowing their positions, calculate the pairwise distance of each galaxy pair to obtain DD distances,

- Apply the same logic but using randomly generated points instead of actual data to calculate RR distances,
- To obtain DR distances, cross-pair the galaxy positions with the random points and calculate the pairwise distances.

Once we have all the pairwise distances, the distribution of DD , DR and RR distances can be laid out with a normalised histogram. The data are binned as needed, and we count the pairs for each bin using `numpy.histogram`. Considering those number counts, the ratio using Equation (2.3) is calculated to derive the two-point correlation function $\xi(r)$.

Random sample: The number of random points does not have to be equal to the number of galaxies. It is best to take a very large number of random points (Landy & Szalay, 1993; Sureshkumar et al., 2021). The measurement of pairwise distances is tested for different random sample sizes to have the required random sample. Small but noticeable differences can be observed in the distributions of distances with lower numbers of random points. However, the distributions seem quite similar for different large sizes of random samples. Hence, we limit the random sample size to 4 times that of the real data.

Periodic boundary conditions: When measuring the pairwise distances, an important assumption of periodicity must be considered. The entire spatial volume of the Universe cannot be simulated such that we could ignore the boundary of the simulation volume. Because of the limited resources, only a representative section of the Universe, a comoving box of size $L = 50, 100, 200 h^{-1}\text{Mpc}$, can be simulated. Particles (or galaxies) close to the boundaries of the box are indeed affected by gravitational forces. Therefore, we have to assume there is something beyond the numerical volume. To that end, the box is extended periodically so that when a particle leaves the box from one side, it will reappear on the opposite side (Mo et al., 2010; Shneider, 2015).

Projected correlation function

The above-mentioned two-point correlation function is the real-space correlation function that does not consider the redshift-space distortions (RSD), which are caused by peculiar velocities of galaxies (section 1.2.2). To include this effect and to compare with observations, the projected correlation function defined by Equation (2.4) (Davis & Peebles, 1983) and measured in redshift space is usually employed.

$$w(r_p) = 2 \int_0^\infty d\pi \xi(r_p, \pi) \quad (2.4)$$

where 2D correlation function $\xi(r_p, \pi)$ is the decomposition of $\xi(r)$ on separations perpendicular (r_p) and parallel (π) to the line of sight. The integration along the line of sight minimises the effects of RSD.

To compute $w(r_p)$, we begin with the calculation of the separations. r_p is the 2D pairwise distances whereas π represents the 1D pairwise distances along the projection axis which is the line of sight. Then, as what was done for the real-space correlation function, the data are binned, and we count the pairs for each bin. In this case, `histogram2d` is the appropriate function because we work on a two dimensional grid. For the integration, `trapz` from `scipy.integrate` is used, which takes the pair counts and the r_p bins as parameters. This process is repeated for all the r_p bins to get all the values of the projected correlation function.

2.3.2 Uncertainty estimation

Estimating errors on clustering statistics plays an important role in constraining cosmological models from galaxy distribution. The optimal way for the uncertainty estimation on a clustering measurement would be to generate a large sample of independent mock catalogues like the data. But this would involve very large N-body simulations.

Instead, the error can be estimated using the data itself. Internal error estimates use a prescription to perturb the data so that copies can be made. These copies allow for an exploration of the data without assuming a specific distribution. This work uses three methods: k-fold, bootstrap and cosmic variance. The first two are generally used for measuring uncertainties in statistical data analysis, while the last one is specifically used in simulations (Norberg et al., 2009; Mo et al., 2010).

k-fold: It consists of splitting the data into k folds or samples ($k = 5$ in our case), and computing the 2pcf for each sample. The uncertainty for each bin is the standard deviation of the k values. It assesses the 2pcf variation when using different subsets of the data.

Bootstrap: A set of n resamplings of the original sample, each having the same size as the original one, is created. The elements of each sample are selected randomly with replacements from the original sample. 2pcf is then computed for each sample, and the uncertainty is standard deviation of the n bootstrap samples for each bin. It is useful when dealing with a limited dataset.

Cosmic variance: The box is split into 8 octants, and the 2pcf is calculated for each octant. The standard deviation of the 8 values for each bin represents the uncertainty. In numerical simulations, it refers to the variance caused by the effects of finite volume, similar to the uncertainty in inferences from observational surveys due to their finite volume (Mo et al., 2010).

Chapter 3

Results and Discussion

3.1 Real-space two-point correlation function

Figure 3.1 displays the real-space clustering measurement of various galaxy samples. We have two samples for each galaxy property. The two-point correlation function $\xi(r)$ is such that r is the projected distance along each axis in the top three rows of panels. The calculated distance for the bottom row of panels is the real 3D distance. The columns show the dependence on galaxy properties: r-i colour, sSFR and age.

The lines from different projection axes are not similar. This is particularly significant for red, quenched and old galaxies where, at small distances, the clustering amplitude is not the same. We note that projections on the y-axis present the lowest amplitude of clustering in general with the projection approach. However, the clustering amplitude of blue, star-forming and young galaxy samples does not change much with the projection axis. At small distances, the values of the measured two-point correlation function are all close to 0.4. Towards larger distances, the clustering amplitude always gets close to 0 for all galaxy samples independently of the choice of the measured distance.

Comparing $\xi(r)$ using real distance to the projection approach, we can see that at small distances, the clustering strength is much higher by a factor of about 4 when using real distance. This is observed with blue, star-forming and young galaxies, which, as mentioned before, looks quite the same with different projection axes. The difference between using projected distance and real distance measurement is even more striking with red, quenched and old galaxies. Consequently, in reality, the clustering amplitude at small scales is undervalued when calculated with projected distance.

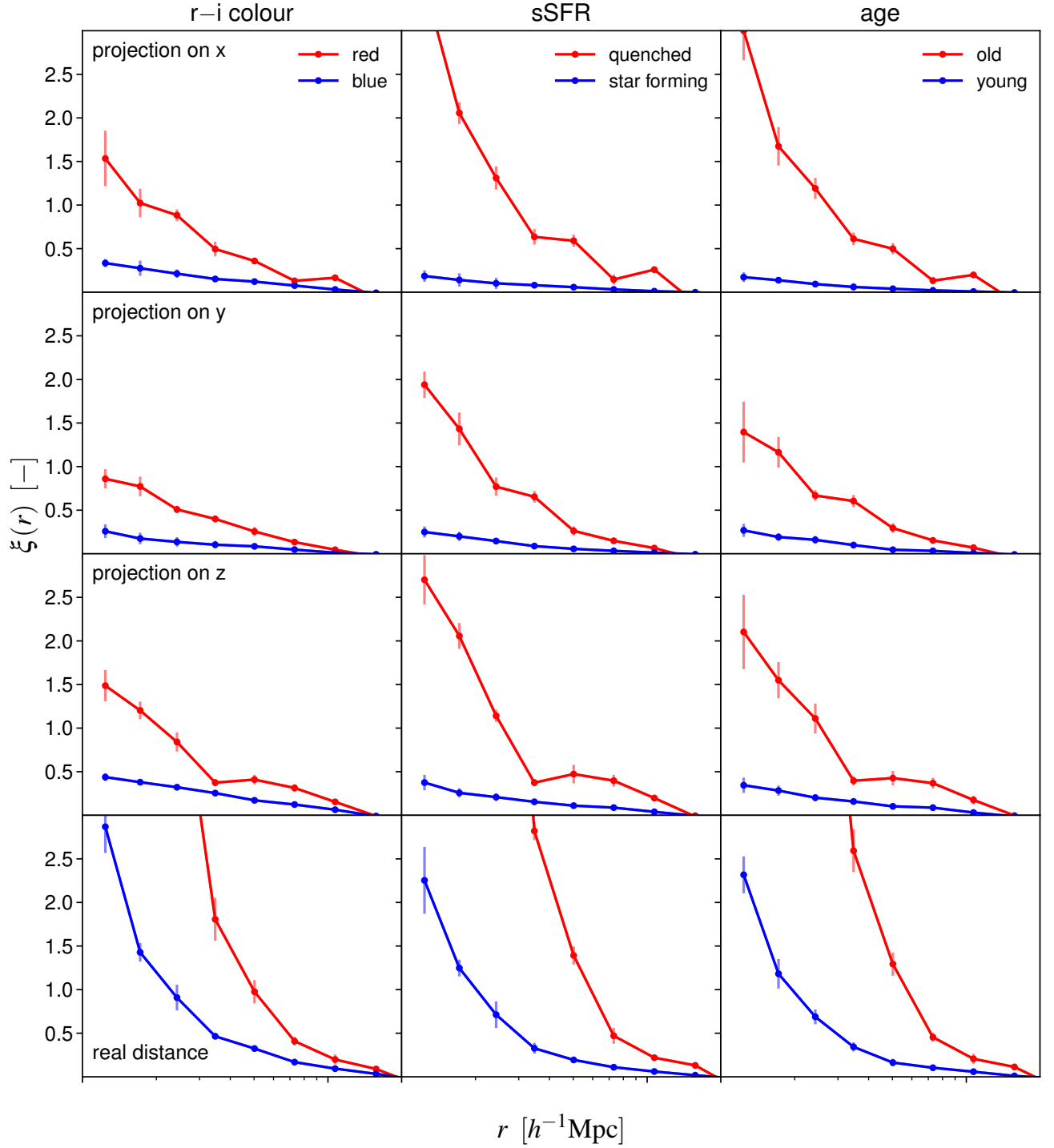


Figure 3.1: Measurement of the two-point correlation function for SIMBA galaxies at $z = 0$ selected by r-i colour, sSFR and age. Each row shows the clustering as a function of the indicated distance (real or projected). The error bars are estimated using the bootstrap method.

A feature consistent throughout every panel is the expected trend that redder, more quenched, and older galaxies cluster more strongly than bluer, star-forming and younger galaxies.

3.2 Resolution analysis

Figure 3.2 shows the change of the projected 2pcf with different resolutions. The sample with the highest resolution is in red, with a minimum resolved galaxy stellar mass of $10^{9.5}M_{\odot}$. The overall shape of the lines remains quite similar with different resolutions. There is a slightly stronger clustering for the sample with a stellar mass resolution limit of $10^{10.5}M_{\odot}$. Hence, we generally have a reasonable resolution convergence.

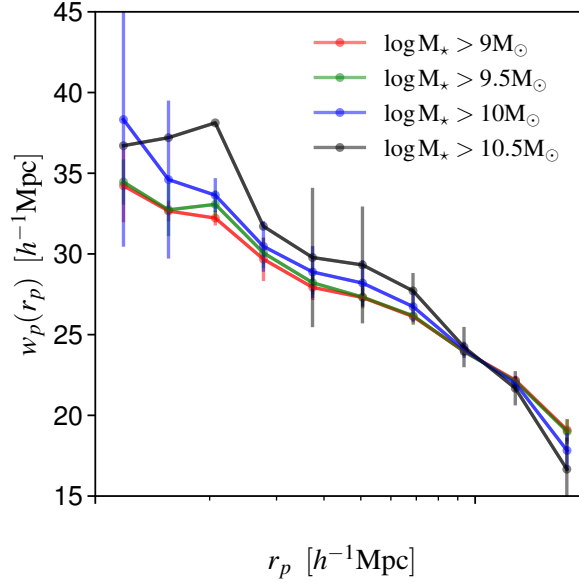


Figure 3.2: Clustering measurement at $z = 0.5$ for four SIMBA galaxy samples with an increasing resolution. Uncertainties are evaluated with the k-fold method.

The clustering is measured for different SIMBA samples applying BOSS probability distribution in Figure 3.3. The lines are almost the same. Therefore, galaxy sample size and random selection of the sample elements do not affect the overall clustering. This implies that the clustering is not sensitive to the number of galaxies in a sample.

3.2.1 Oversampling

New galaxy positions are obtained after adding random values to the positions the interpolation gives. As mentioned in section 2.2.1, those values are drawn from a Gaussian distribution centred on 0 with a standard deviation δ , where δ is the average distance between the galaxies.

Table 3.1 shows p-values computed from two statistical tests: Student's t-test and Kolmogorov-Smirnov test. p-value is calculated for each coordinate x, y, z . Those values inform whether the new positions with random values are still drawn from the same distribution as the original ones.

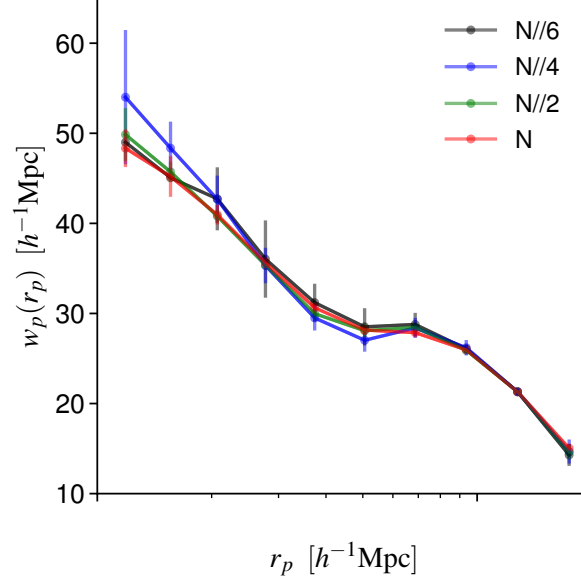


Figure 3.3: Clustering measurement at $z = 0.5$ for four SIMBA galaxy samples of different sizes using BOSS probability distribution applied to SIMBA. Uncertainties are evaluated with the k-fold method.

We use a significant level of 0.05, the typical threshold value, to compare the result. A p-value less than 0.05 would confirm that the samples are different, the opposite being not necessarily true. Thus, the choice should be a higher p-value. Nevertheless, the random spread should not be too high since, in that case, the newly added galaxies would fill up voids in between haloes.

All the tests we performed to find the adequate value of δ are shown in Table B.1 of Appendix B. We start by varying δ from the size of the simulation box divided by the number of galaxies, to the size of the box divided by two in logarithmic steps and compute p-values corresponding to each δ . At first sight, we see that the adequate δ should lay between 0.025 to 0.25. Beyond that range, p-values are either too high or too low, resulting in unreliable clustering measurements as seen in Figure A.3 of Appendix A. Table 3.1 shows p-values for relevant δ within that range. The distributions of the positions, displayed in Figure 3.4, do not show any noticeable difference. The p-values are not all the same for each coordinate. We have to find the right balance so that the shape of the correlation function line, as shown in Figure 3.5, is still preserved. Hence, we choose $\delta = 0.04$.

Table 3.1: p-values corresponding to $\delta = 0.03 - 0.04 - 0.05 - 0.06$ comparing new positions (x, y, z) with the original ones using t-test and KS test.

	δ	x	y	z
t-test	0.03	0.78339539	0.54704500	0.38707646
	0.04	0.90730394	0.40723920	0.23105650
	0.05	0.83624541	0.33280744	0.14427096
	0.06	0.90822086	0.23367908	0.05364848
KS test	0.03	0.00265565	0.07722337	0.15871622
	0.04	0.00151987	0.03431718	0.07722337
	0.05	0.00114009	0.03081220	0.02473465
	0.06	0.00175113	0.02762617	0.00667843

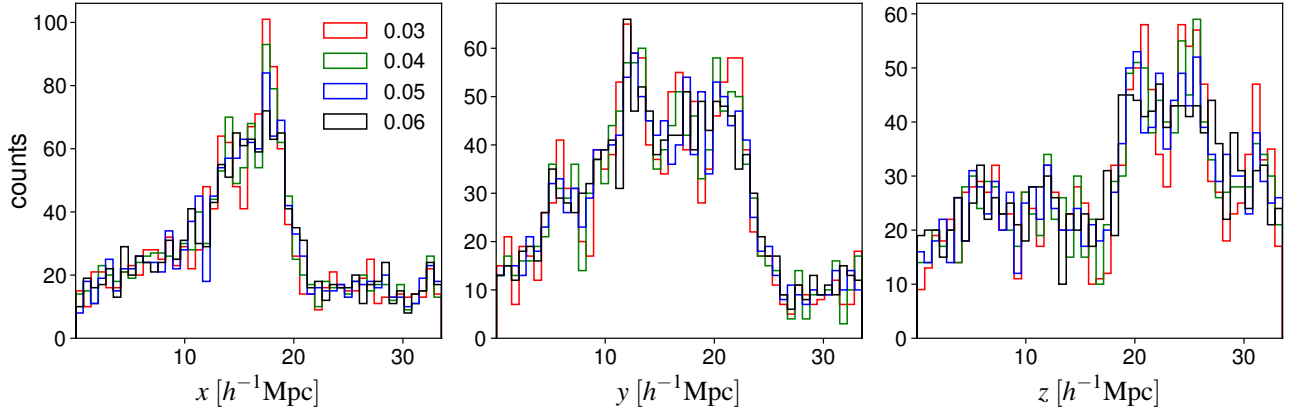


Figure 3.4: Distributions of galaxy positions for four samples with different δ as indicated.

Regarding statistical significance, the p-values from $\delta = 0.04$ indicate that the observed difference between the two distributions is not statistically significant. Therefore, adding the random values (drawn from a Gaussian distribution centred on 0 with a standard deviation $\delta = 0.04$) to the original positions leads to the conclusion that there is insufficient evidence to say the two distributions of the positions are different. Using this value of δ would give reliable clustering measurements.

In Figure A.2 of Appendix A, other alternatives were tested. We tried adding random values not drawn from a normal distribution. The resulting distribution of the positions is however definitely not similar to the original ones.

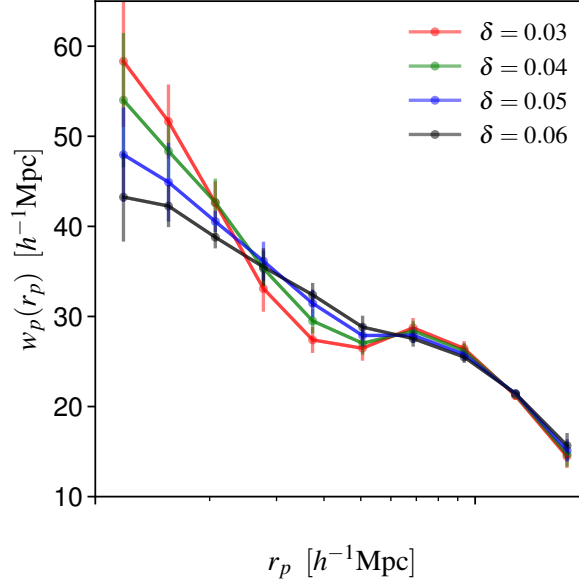


Figure 3.5: Projected correlation function measured for four samples with different δ as indicated. k-fold is used to compute the error bars.

Randomising galaxy positions after the interpolation aimed to prevent strong clustering signal at the lowest distances, which would lead to implicit clustering creating unwanted bumps in the correlation function. However, a bump still appears at around $10 h^{-1}\text{Mpc}$ as seen in Figure 3.3 and Figure 3.5. It could be due to halo clustering, with the value $10 h^{-1}\text{Mpc}$ indicating the average distance between haloes.

3.2.2 Matching data by replicating the simulation box

Figure 3.6 compares the mass distributions of the BOSS sample, SIMBA galaxies from the original box, and from 9 and 10 times replications on each side of the box. We assess two mass ranges where the observed peak of the stellar mass distribution is located. The first mass range is $11.05 < \log M_{\star}/M_{\odot} < 11.55$, while the second range is broader, $10.95 < \log M_{\star}/M_{\odot} < 11.75$.

The narrower mass range where the observed peak is ($11.05 < \log M_{\star}/M_{\odot} < 11.55$), shows that 10 replications of the box size would give an almost similar mass peak as that of the BOSS sample. However, the other range (where the BOSS peak is not located) has many galaxies that would likely impact the clustering measurements. An even higher number of those galaxies are obtained with 9 replications of the box size. As for the broader mass range $10.95 < \log M_{\star}/M_{\odot} < 11.75$, 10 replications result in an excessive number of galaxies within that mass range, even though the number of galaxies beyond that range is relatively low. By replicating the box size 9 times, we

have a reasonable number of galaxies within that mass range $10.95 < \log M_*/M_\odot < 11.75$ where the peak is. Beyond that range, there are fewer galaxies than we got from the narrower mass range $11.05 < \log M_*/M_\odot < 11.55$.

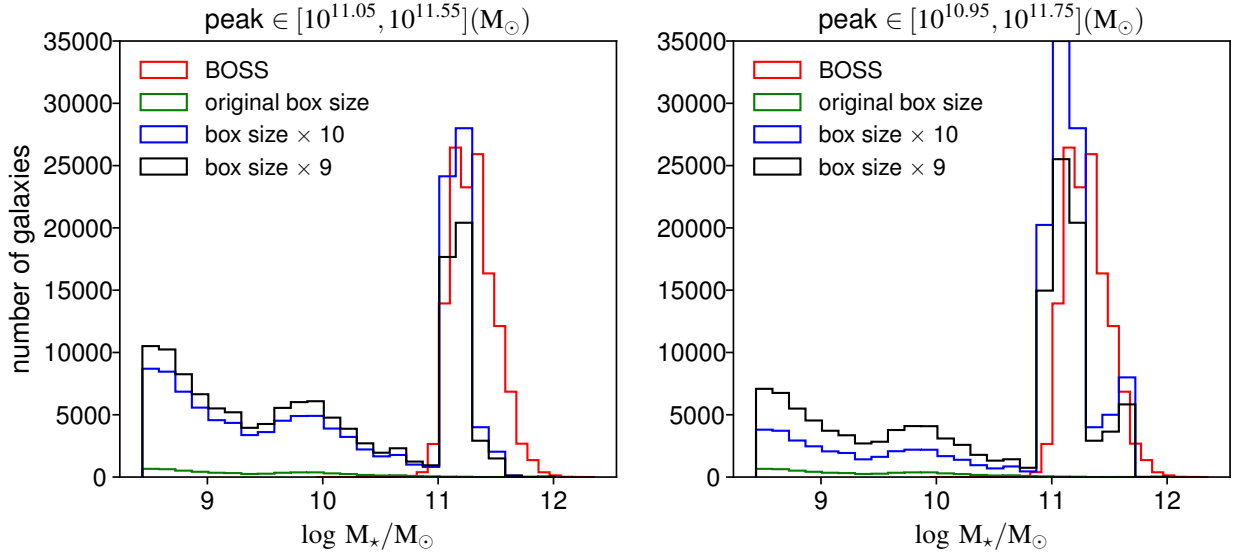


Figure 3.6: Comparison of stellar mass distributions of BOSS and SIMBA galaxies. The smaller mass range where the peak is located, $11.05 < \log M_*/M_\odot < 11.55$, is on the left, and the broader one $10.95 < \log M_*/M_\odot < 11.75$ on the right. SIMBA galaxies from the original simulation box are in green, while the 9 and 10 times replicated box sizes are in black and blue, respectively.

The density of galaxies, after applying 9 replications of the size of the simulation box, is shown in Figure 3.7. Before randomly selecting galaxies in the mass range where the observed peak is, we can clearly see a repetitive pattern that reflects the replication of the simulation box. After the selection, the density is distinctly lower.

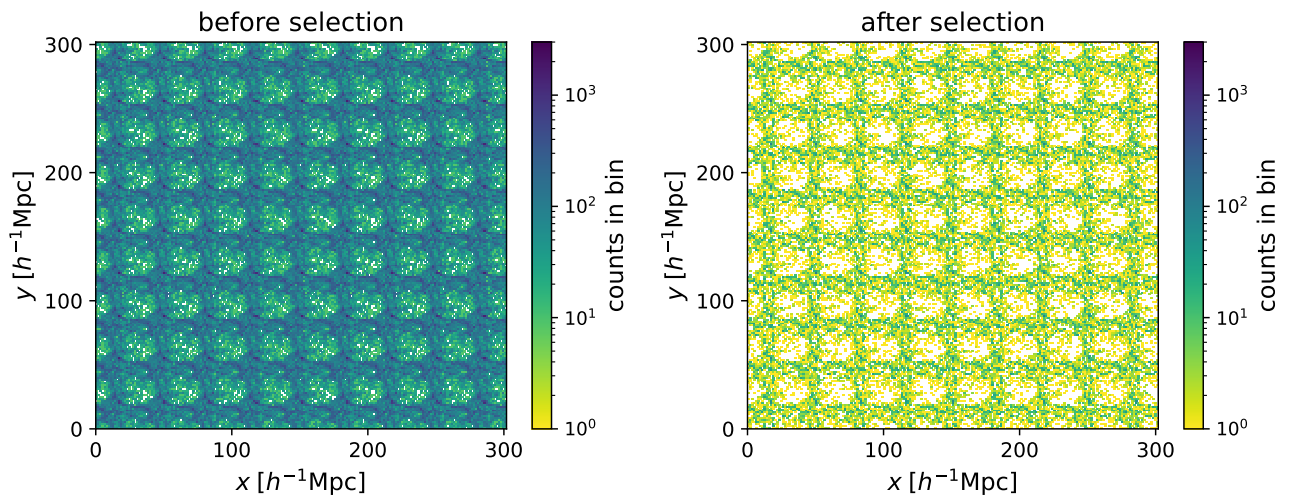


Figure 3.7: Galaxy density following 9 replications of the original size of the simulation box before (*left*) and after (*right*) random selection of galaxies beyond the mass range $10.95 < \log M_*/M_\odot < 11.75$

The problem with this replication approach is the new distributions. Even if there is a reasonable number of galaxies in the mass range $10.95 < \log M_*/M_\odot < 11.75$, there are still many low-mass galaxies. We also end up with many haloes not hosting any galaxy. With a total number of haloes of 61 977 in the original SIMBA simulation box, 58 239 contain galaxies and 3 738 do not. Replications of the simulation box would give too many haloes without galaxies. Moreover, Figure 3.7 reveals the density is homogeneous, which does not reflect real data.

3.3 Projected correlation function: comparison with literature

The measured projected correlation functions $w_p(r_p)$ for different galaxy samples are shown in Figures 3.8, 3.9 and 3.10. The $w_p(r_p)$ calculation takes the z-axis as the line of sight. The results detailed in the section 3.2.1 are taken into account. Therefore, random values drawn from a Gaussian distribution, centred on 0 with a standard deviation of 0.04, are added to the positions before computing the projected 2pcf.

Dependence on colour: Zehavi et al. (2011) analysed galaxy clustering in the SDSS DR7 and the dependence on colour. Various galaxy subsamples selected by colour from a fixed volume-limited sample determined by luminosity are used to assess the dependence. In Figure 3.8, we notice that the clustering amplitude of redder galaxies is higher than that of bluer galaxies. The differences in clustering between the red and blue samples are stable at small separations before getting weaker from $\sim 2 h^{-1}\text{Mpc}$. These behaviours are also seen in Zehavi et al. (2011) with different methods and data employed for the measurements. In our result, we can observe the clustering dependence up to separations of $\sim 3 h^{-1}\text{Mpc}$. However, the dependence is no longer noticeable at larger separations, including where the previously mentioned bump appears.

Dependence on sSFR: Here, we refer to the clustering dependence on sSFR from Mostek et al. (2013) where DEEP2 data are used. Figure 3.9 follows the general expectation that quenched galaxies (lower sSFR) are more clustered than star-forming ones (higher sSFR). However, because the selection of samples and the data differ, the amplitude and the shape are not similar. We find that the slope of the correlation function is steeper for quenched galaxies than that of the star-forming sample. As for the colour-dependent analysis, the clustering dependence on sSFR is only important at small distances up to $\sim 1.5 h^{-1}\text{Mpc}$.

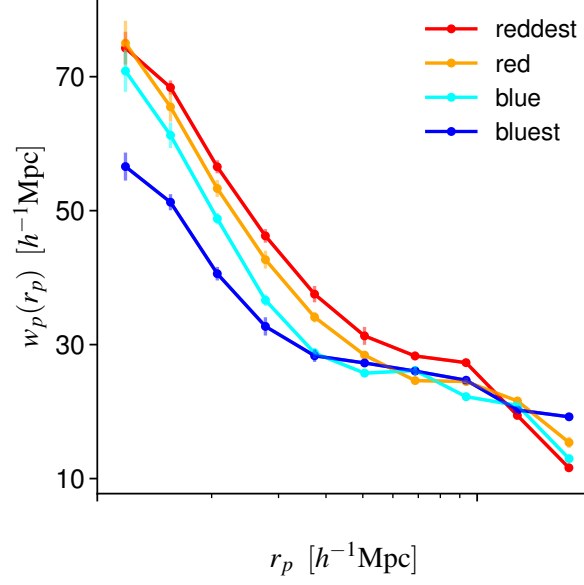


Figure 3.8: Projected correlation function for SIMBA galaxies selected by $r-i$ colour. The error bars are computed from five bootstrap resamplings.

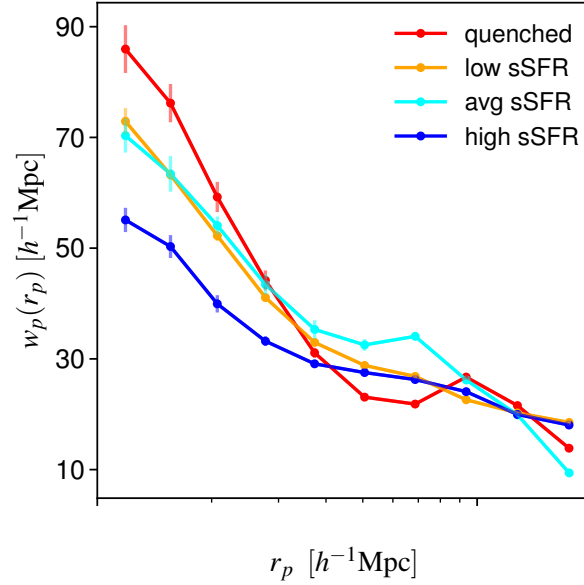


Figure 3.9: Projected correlation function for SIMBA galaxies selected by $sSFR$. The error bars are computed similarly to Figure 3.8.

Dependence on age: The clustering dependence on age shown in Figure 3.10 is less significant compared with the results found in Zhai et al. (2023). However, this could likely be due to the choice of data that we used for the interpolation. The right panel in Figure A.1 of Appendix A, indeed, shows a more significant age dependence. We also find that the differences in clustering from old and young galaxy samples seem less evident compared with that from red and blue, quenched and star-forming samples. But once again, if we look at Figure A.1, they are almost similar for all three properties.

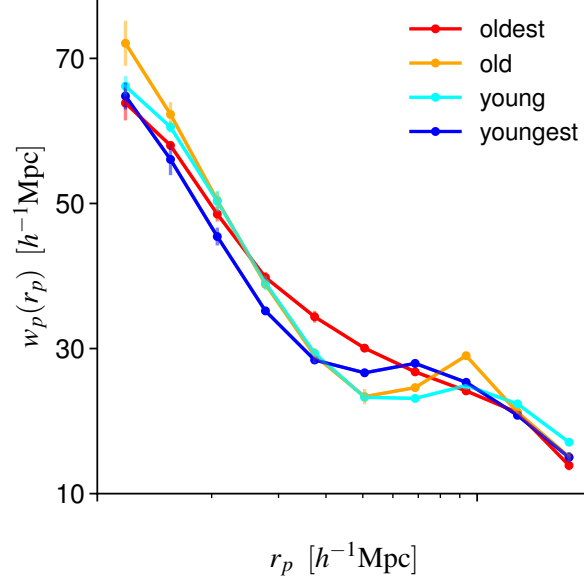


Figure 3.10: Same as Figure 3.8 and Figure 3.9 but for age-dependent analysis.

The statistical results suggest that the clustering measurements after oversampling are reliable. But even after the interpolation, SIMBA and BOSS data are likely still poorly matched, given that the overall shape of the clustering measurements is different.

Additionally, the powers from BOSS correlation functions are much higher than we get from SIMBA. This is in part because the observation sample is based on the redshift band $z = 0.45 - 0.55$ while with SIMBA, we only have a box of $\sim 33.5 h^{-1} \text{Mpc}$ at $z = 0.5$. A multiplicative factor related to the equivalent distance for the redshift band would need to be included to match the BOSS sample better. That would deepen the simulation volume and increase the number of galaxies. However, implementing this would be an expensive calculation.

Regarding the sample selection, we proceeded more simply by splitting the initial full sample into four subsamples as a function of relevant galaxy properties. A similar method is applied in Artale et al. (2017) with different subsample sizes. This differs from the methodology in Zhai et al. (2023) which is more complex since fixed number densities are used as thresholds to create the subsamples. Moreover, there is also the difference in the sky coverage between observation and simulation, with the latter only having the size of the simulation box.

To get the exact values of $w_p(r_p)$ in Zhai et al. (2023) using Equation (2.4), we would need to mimic all observational setups. That would require a much larger simulation volume.

Regardless, the clustering measurements as a function of each galaxy property generally follow the same direction as seen in Zhai et al. (2023) on projected scales up to $r_p \sim 3 h^{-1} \text{Mpc}$.

In all three analyses (Figures 3.8, 3.9 and 3.10), the clustering dependence on each property is observed at small separations only. The comoving size of the simulation box ($50 h^{-1}\text{Mpc}$) could be the reason for that. Smaller boxes cannot produce the largest cosmic structures. Hence, it is related to cosmic variance. Because of the limited volume, the clustering may be underpredicted at the high end of separations. The estimated uncertainties using the same box are also low at these scales. So to assess the impact of the simulation volume, we need to measure the clustering in other simulation boxes of different sizes.

3.4 Comparison using simulation boxes of different sizes

The clustering measurements in section 3.3 use galaxies from the $50 h^{-1}\text{Mpc}$ simulation box. Figure 3.11 shows the correlation functions using two other boxes of different sizes, $25 h^{-1}\text{Mpc}$ and $100 h^{-1}\text{Mpc}$. Previous measurements from the $50 h^{-1}\text{Mpc}$ volume are also shown for comparison.

The powers of the signals from the smaller box, $25 h^{-1}\text{Mpc}$, vary from about 0 to 40, whereas the larger box $100 h^{-1}\text{Mpc}$, reveals higher w_p around 45 to 75. On the other hand, the power range from the $50 h^{-1}\text{Mpc}$ box is broader, from 10 to 80.

With the $25 h^{-1}\text{Mpc}$ box, we can hardly see the expected trends, likely owing to the small size of the box. We note a bump at $\sim 5 h^{-1}\text{Mpc}$ (a dip at $\sim 2 h^{-1}\text{Mpc}$), which is more pronounced than the one at $\sim 10 h^{-1}\text{Mpc}$ with the $50 h^{-1}\text{Mpc}$ box. We suggested that the latter could be the result of halo clustering. In this case, many massive large haloes exist, according to the halo mass distribution on the left panel of Figure 3.12. This might contribute to the galaxy clustering amplitude which is higher around this $r_p \sim 5 h^{-1}\text{Mpc}$. However, we do not exclude the possibility that it could be the result of some periodic boundary effect.

For the $100 h^{-1}\text{Mpc}$ box, the clustering as a function of colour shows that there is no real visible trend. The curves of the reddest and bluest galaxy samples are in between the red and the blue samples curves. As for the dependence on age, the oldest galaxy sample differs from the rest of the samples. Only the analysis based on sSFR presents the expected result that galaxies with a high sSFR are less clustered than those with a low sSFR. However, the clustering amplitude of the sample with the lowest sSFR is lower than expected. Though the resolution is the same as with the $50 h^{-1}\text{Mpc}$ box, the selection of galaxies based on a given property creates different samples, hence different clustering behaviour, in this $100 h^{-1}\text{Mpc}$ box.

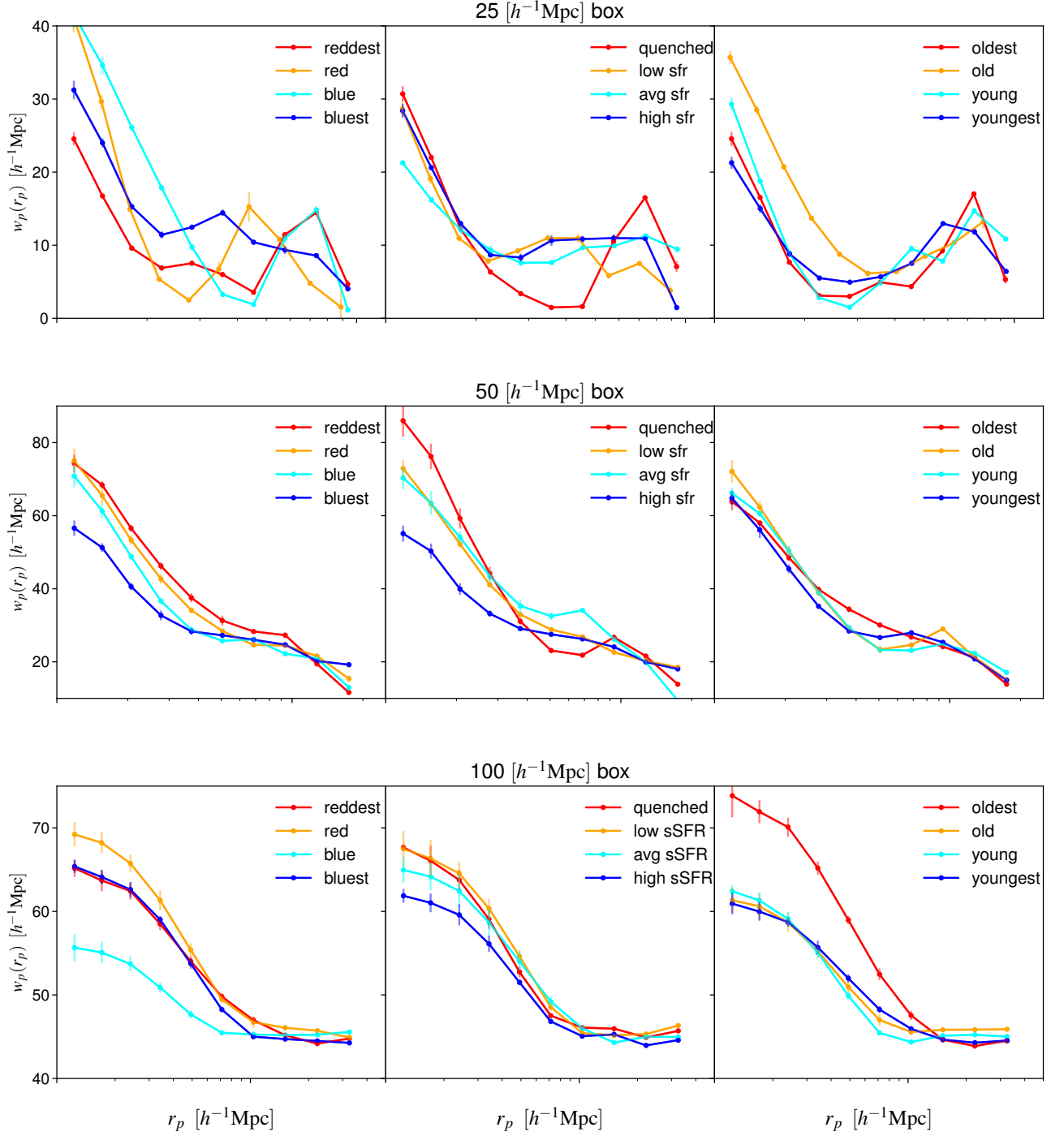


Figure 3.11: Measurement of the correlation function $w_p(r_p)$ for SIMBA galaxies selected by $r-i$ colour (*left panels*), sSFR (*middle panels*) and age (*right panels*), from three different simulation boxes.

There is no bump (or dip) at $r_p > 10 h^{-1}\text{Mpc}$, where we would expect to see it based on the trends in the smaller boxes. This is because the size of the volume allows for the presence of more massive haloes, leading to more signals around where a dip would have appeared. The right panel of Figure 3.12 shows that the $100 h^{-1}\text{Mpc}$ has, indeed, a broader halo mass range, with more haloes on the

massive end. These confirm that larger structures produced in this volume affect the overall galaxy clustering patterns in different environments.

Therefore, even if the influence of cosmic variance is reduced with the $100 h^{-1}\text{Mpc}$ box, some trends might become less noticeable.

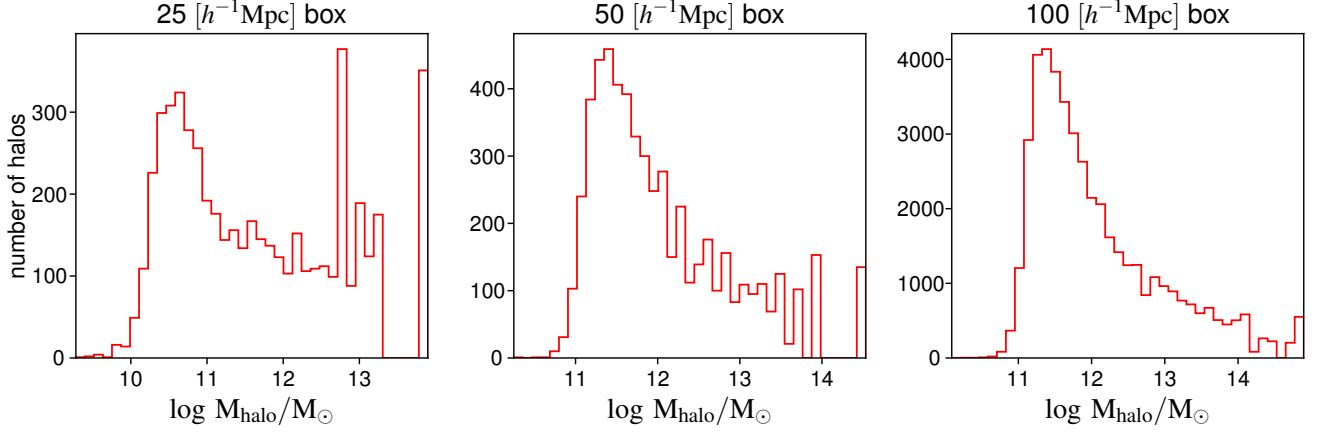


Figure 3.12: Halo mass distribution for the three SIMBA simulation boxes at $z = 0.5$: $25 h^{-1}\text{Mpc}$ (left), $50 h^{-1}\text{Mpc}$ (middle) and $100 h^{-1}\text{Mpc}$ (right).

Effect of the presence of large massive haloes

Figure 3.12 indicates there are more larger haloes in the $100 h^{-1}\text{Mpc}$ box. In Figure 3.13, we repeat the previous clustering measurements but exclude the most massive haloes ($M_{\text{halo}} > 10^{13.5} M_{\odot}$). We can see that signals from blue, star-forming and young galaxies are now higher. The excluded massive haloes likely host more red, quenched and old galaxies. Their exclusion leads to higher clustering amplitude of the remaining galaxy samples at small scales.

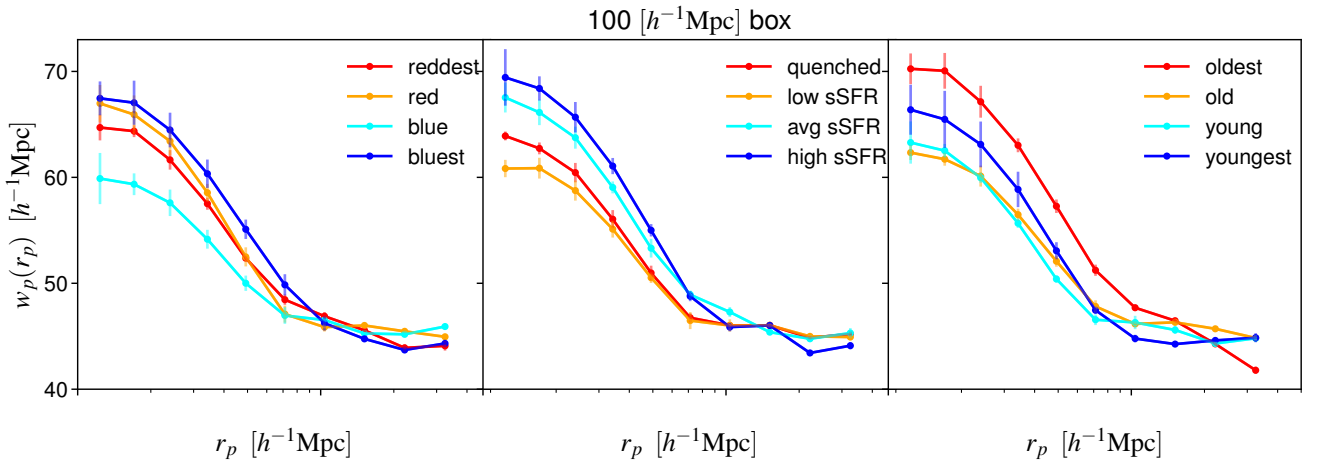


Figure 3.13: Measurement of the correlation function $w_p(r_p)$ for SIMBA galaxies selected by $r-i$ colour (left), $s\text{SFR}$ (middle) and age (right), in a simulation volume of $100 h^{-1}\text{Mpc}$. Galaxies in heaviest haloes ($M_{\text{halo}} > 10^{13.5} M_{\odot}$) are excluded.

We also note a minor change at $11 h^{-1}\text{Mpc}$; the curves seem to show a very small bump for some samples. These curves look almost flat at scales larger than $10 h^{-1}\text{Mpc}$ when massive haloes are included (lower panels of Figure 3.11). We can then conclude that the bump (or dip), which was already mentioned multiple times, is due to halo clustering.

The clustering should be studied on even larger volumes to obtain more realistic results in good agreement with observation. The clustering analysis with EAGLE simulation also mentions this limitation (Artale et al., 2017). They concluded that a volume of $67.77 h^{-1}\text{Mpc}$ is too small, so their study focused on scales smaller than a few Mpc. Their clustering measurements as a function of stellar mass, colour and luminosity agree with results from the GAMA survey. In Springel et al. (2018) with IllustrisTNG, galaxy clustering was measured in a simulation volume ($205 h^{-1}\text{Mpc}$) twice as big as the largest box of SIMBA. The 2pcf as a function of stellar mass and galaxy colour of the simulated galaxies agrees well with SDSS data. These highlight the importance of having a large simulation volume to analyse galaxy clustering.

The main findings indicate that redder, older and more quenched galaxies cluster more strongly than bluer, younger and star-forming ones, whether in real space or redshift space. These differences in clustering should reflect different environments where galaxies reside and infer the properties of the host dark matter haloes. Understanding the clustering behaviour of the galaxies and its dependence on galaxy properties can be used to trace the cosmic structure.

The clustering measurements in real space imply that without additional data processing or trying to match with observation, theoretical estimation gives the expected trends. And the 2pcf measured in redshift space only confirms those results at small distances.

We can conclude that high-density environments in the simulation are more likely to be populated by redder, quenched and older galaxies, which is consistent with the literature (Zehavi et al., 2011; Mostek et al., 2013; Artale et al., 2017; Zhai et al., 2023). Therefore, SIMBA generally reproduces galaxy clustering based on galaxy properties but only at small scales up to $3 h^{-1}\text{Mpc}$ due to the limited size of the simulation volume.

Conclusion

This thesis utilised the SIMBA cosmological hydrodynamical simulation to predict and analyse the clustering characteristics of galaxies. The aim was to explore the intricate relationship between galaxy properties and their environmental influences. Examining the clustering properties across various simulations and observational datasets was crucial in revealing how galaxy properties correlate with the density of their environments, reflecting the underlying dark matter distribution. We computed the two-point correlation functions of various galaxy samples selected by colour, specific star formation rate, and age, by referring to previous studies. Our work has provided precious insights into how galaxies evolve in different cosmic environments through rigorous statistical analysis and comparison of simulations outputs with observational data.

It has shown that redder, more quenched, and older galaxies are typically found in denser environments, influencing their clustering behaviour.

Our main findings are detailed as follows.

- (i) In real space, redder, more quenched and older galaxies present a higher clustering. The clustering amplitude is undervalued when estimated with the projected distance instead of the three-dimensional distance.
- (ii) Clustering measurements of SIMBA galaxy samples with different resolutions indicate a reasonable convergence. The clustering is also not sensitive to the number of galaxies and the random selection of elements in a sample.
- (iii) A matching of the simulation and observational data based on probability distribution provides similar stellar mass distributions. Random values, drawn from a Gaussian distribution centred on 0 with a standard deviation of $\delta = 0.04$, were added to the positions of SIMBA galaxies. The resulting clustering measurements are reliable. Alternatively, a simple replication of the simulation box to obtain the number of galaxies in BOSS, provides unreliable distributions
- (iv) The projected correlation function shows that redder, more quenched and older galaxies tend to cluster more strongly, consistent with the literature. Because of the limited simulation vol-

ume, the clustering dependence on each property is only seen at small scales up to $3 h^{-1}\text{Mpc}$. Additional data treatments are required to match observations better and have the power of the correlation functions from these observational datasets.

- (v) The size of the simulation volume has a significant impact on the clustering measurements. The smaller box of $25 h^{-1}\text{Mpc}$ hardly gives the expected clustering trends. Halo clustering affects the power of the signals, especially with the $100 h^{-1}\text{Mpc}$ box. The trends in the $50 h^{-1}\text{Mpc}$ box are less pronounced with the larger box. More accurate measurements would require a larger simulation volume.

Measurements of galaxy clustering as a function of colour, star formation, age or other parameters provide valuable information to constrain galaxy formation and evolution models. Our findings confirm existing theories about how galaxies form and evolve, and highlight the importance of considering environmental effects. They contribute to our understanding of the interplay between galaxy properties and the large-scale structure, and the role of dark matter in shaping the Universe.

Future work would involve refining these models further, focusing on a deeper exploration of the galaxy-halo connection. We could also incorporate a broader range of galaxy properties, and explore their implications for cosmology.

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Appendix A

Interpolation and oversampling trials

In this appendix, we report the results of the trials that were performed regarding the interpolation (section 2.2.1) and the oversampling (section 3.2.1). Figure A.1 shows the clustering measurement as presented in section 3.3, but with different data. SIMBA's data (at $z = 0.55$) are matched with BOSS data based on the stellar mass function presented in Maraston et al. (2013) in the redshift bin $z = 0.5 - 0.6$ (this bin also falls within the range that is used in Zhai et al. (2023) $z = 0.48 - 0.62$). Compared with Figures 3.8, 3.9 and 3.10, the correlation functions do not display any stark difference. The overall shapes are similar. An apparent change can be noticed in the age-dependent analysis where the clustering of older galaxies seemed to be more stronger with these other data.

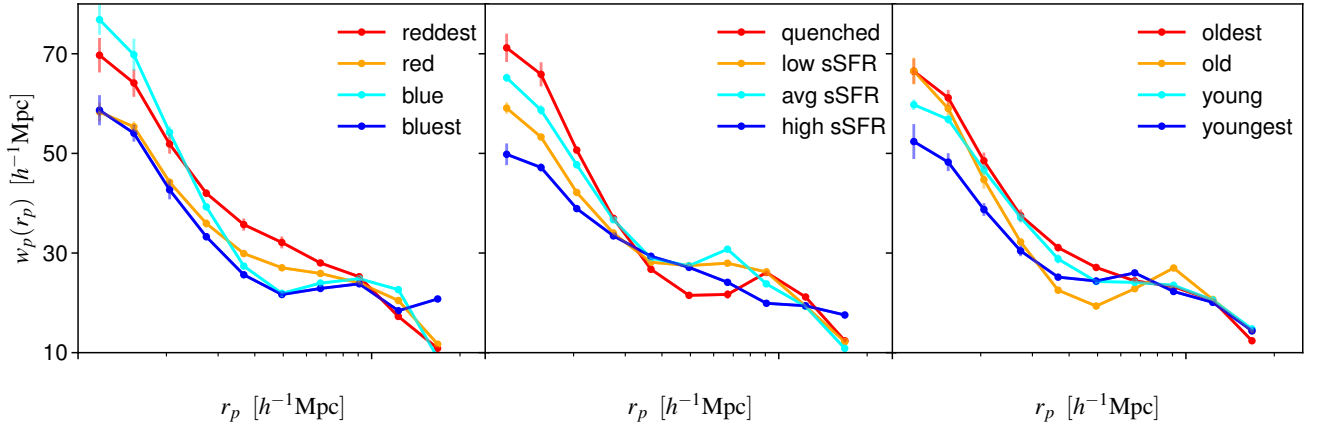


Figure A.1: Projected correlation function for SIMBA galaxy samples selected by $r-i$ colour, sSFR and age at $z = 0.55$ after interpolation following the stellar mass function of BOSS galaxies within the redshift range of $z = 0.5 - 0.6$. Uncertainties are from five bootstrap resamplings.

During the data analysis, the oversampling required to add random values drawn from a Gaussian (normal) distribution centred on 0 with a standard deviation of $\delta = 0.04$. The oversampling with values from different distributions are shown in Figure A.2 and compared to the original values.

We tested with:

- gaussian distribution centred on 0 with a standard deviation of δ where δ is the size of the simulation box divided by size of the galaxy sample.
- uniform distribution
- random distribution in the half-open interval $[0.0 - 1.0)$

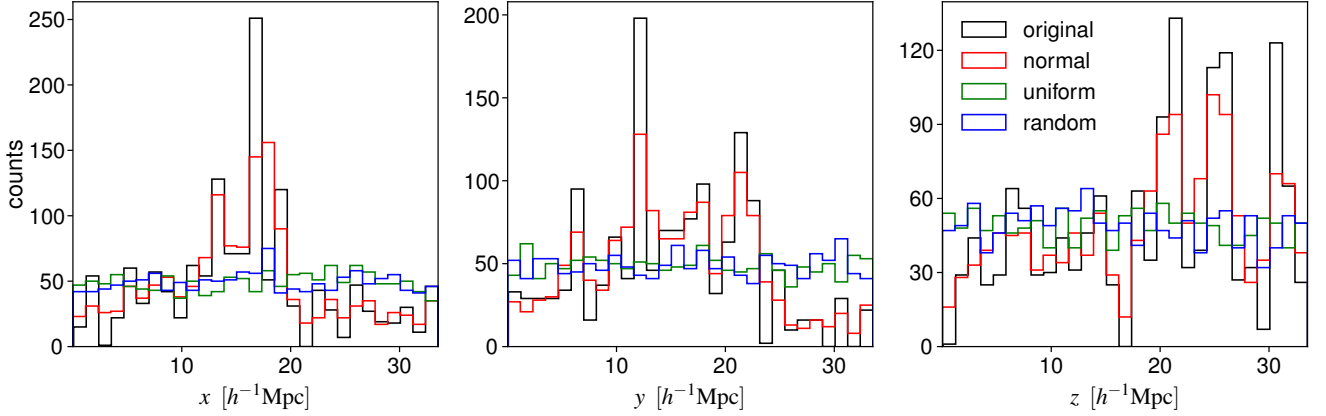


Figure A.2: Distributions of the positions before (black) and after adding random values. Those values are drawn from normal (red), uniform (green) and random (blue) distributions.

Figure A.3 shows the projected correlation functions for $\delta = 0.00025 - 0.0025 - 0.025 - 0.25 - 2.5 - 25$. We can see that with $\delta > 0.25$, we have almost flat lines, whereas with $\delta < 0.025$, the lines have a shape with bumps and valleys.

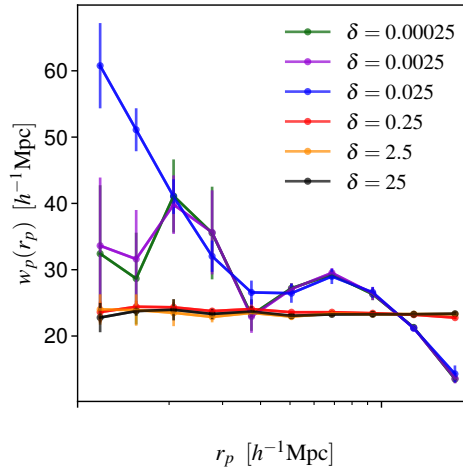


Figure A.3: Clustering measurements for $\delta = 0.00025 - 0.0025 - 0.025 - 0.25 - 2.5 - 25$. The uncertainties are measured with the k-fold method.

Appendix B

Statistical analysis: p-values

Table B.1: p-values corresponding to δ between 0.00025 to 25 comparing new positions (x,y,z) with the original ones using t-test and KS test.

	δ	x	y	z
t-test	0.00025	0.99983240	0.99944384	0.99890422
	0.0025	0.99832446	0.67240458	0.71534604
	0.025	0.85119491	0.53751131	0.48514887
	0.03	0.78339539	0.54704500	0.38707646
	0.04	0.90730394	0.40723920	0.23105650
	0.05	0.83624541	0.33280744	0.14427096
	0.06	0.90822086	0.23367908	0.05364848
	0.08	0.75492133	0.19597246	0.01176533
	0.10	0.93479476	0.10168043	0.00151300
	0.12	0.98671123	0.04022337	0.00134834
	0.25	0.01780074	$1.31232729 \times 10^{-05}$	$7.58361472 \times 10^{-11}$
	2.5	0.00048866	$3.68771183 \times 10^{-07}$	$3.62356095 \times 10^{-14}$
	25	$1.70425123 \times 10^{-05}$	$2.15094801 \times 10^{-06}$	$6.90950293 \times 10^{-14}$
KS test	0.00025	0.90960558	0.78201663	0.99568740
	0.0025	0.78201663	0.81086741	0.62680168
	0.025	0.00397647	0.06358728	0.21983696
	0.03	0.00265565	0.07722337	0.15871622
	0.04	0.00151987	0.03431718	0.07722337
	0.05	0.00114009	0.03081220	0.02473465
	0.06	0.00175113	0.02762617	0.00667843
	0.08	0.00063072	0.00304248	0.00063072
	0.10	$3.95733240 \times 10^{-05}$	0.00024874	$1.62142252 \times 10^{-05}$
	0.12	$5.30342004 \times 10^{-06}$	$5.30342004 \times 10^{-06}$	$2.97552062 \times 10^{-06}$
	0.25	$3.39827409 \times 10^{-21}$	$4.91609186 \times 10^{-21}$	$2.08210656 \times 10^{-15}$
	2.5	$7.03038506 \times 10^{-29}$	$2.16693198 \times 10^{-27}$	$4.37041766 \times 10^{-20}$
	25	$2.17692962 \times 10^{-33}$	$4.54979149 \times 10^{-29}$	$2.12201190 \times 10^{-20}$

Appendix C

Stellar mass distributions

Table C.1: Stellar mass distribution for BOSS galaxies within the range $0.45 < z < 0.55$ (Maraston et al., 2013) and SIMBA galaxies at $z = 0.5$

$\log M_{\star}/M_{\odot}$	BOSS	SIMBA	$\log M_{\star}/M_{\odot}$	BOSS	SIMBA
9.95	2	245	11.15	26446	23
10.05	7	187	11.25	23258	7
10.15	2	171	11.35	25919	4
10.25	4	149	11.45	16346	1
10.35	6	106	11.55	12119	5
10.45	10	82	11.65	6851	6
10.55	18	93	11.75	2678	1
10.65	37	89	11.85	1361	1
10.75	93	48	11.95	383	1
10.85	391	46	12.05	134	0
10.95	2656	31	12.15	22	0
11.05	13936	23	12.25	10	0